The a_0 Law with Cold Dark Energy Counted Once: the Form the Record Closes On, Its Gates, and What Stays Open

Working note (PAPER72), version 1.3, 2026-11-13 (v1.2: DOI 10.5281/zenodo.24182757; v1.1: DOI 10.5281/zenodo.24125791). AI-assisted research programme; not peer reviewed; not reproduced by any outside group.
Companions: PAPER69 (DOI 10.5281/zenodo.24025170, the closure record), PAPER70 (v2.1, DOI 10.5281/zenodo.24092073, field equations and the record since) and PAPER71 (v1.1, DOI 10.5281/zenodo.24095262, the switch class). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

This note states the form on which the record closes and grades it. It does not complete the theory. The a_0 law alone fails the CMB, the growth of structure, clusters and the Bullet, each for want of the same cold, clustering component of about 5.4 times the baryons. Every way tried of keeping that component out of galaxies has a failing verdict; the last one is recorded here. The form chosen by the owner on 2026-11-09 counts the component once: it is in galaxies too, and around condensed baryons the law describes how it sits. It is real mass, not a second force. In the adopted action this is the static state's own reading: the law's halo is the khronon sector's energy density, carried by the one conserved charge that is dust in the expanding background. We regrade the form with criteria frozen before any number. On groups and clusters it is not falsified, by a gate that can fail from one side only, and the richest clusters sit at its ceiling; the Bullet passes only if the subcluster's stripped gas is counted. On hosted dwarfs it fails: it removes most of the law's shortfall in every class, but two classes stay high and four low rows cannot be reached. On KiDS lensing by galaxy type the test returns no verdict; its readings put early types between the law and the form's truncated halo and cannot place discs; on host-free lenses the form passes a gate that does not separate it from the law. The form passes rotation curves. Its CMB, lensing and growth rows hold when computed, on an amended reading of that lane's checks. With the amplitude of fluctuations pinned by large-angle polarisation, CMB lensing matches the prediction to 0.3σ and the growth rate passes. Raw KiDS weak lensing, on a gravity-only spectrum, prefers an amplitude five of its own widths lower: the absolute fit is excluded at 3σ by the frozen line and reads tension once the reference run's realisation noise is corrected, a check made after grading. This is the tension that any cold component fitted to the CMB carries; the form states neither gas ejection nor neutrino mass, and does not relieve it. With gravity coupled the static state is linearly stable around a point mass; an instability found earlier belonged to a planar background.

Version 1.1 adds the lanes that landed on 2026-11-10 and states the theory in six rules (Section 7). The collapse was computed and breaks down: a clock that starts as dust folds at the centre, and the equations give no answer beyond that moment. The static state is stable around isolated discs away from their centres; a disc in an external field has no verdict, with stability excluded there, and a disc's own centre is open. No dust term tried, keyed on the clock's offset, on its expansion or on its acceleration, joins the falling state to the standing one; the clock's leafwise flux cannot supply the halo; and the action has no cluster state at the form's own dust constant. A survey of media with more structure than one clock finds no class without a failing verdict. On the data side, most of the weak-lensing gap was the survey's superseded release and our own inputs: on KiDS-Legacy, with a neutrino floor, the covariance at the model and the reference run's draw corrected, the form's amplitude sits 1.71/1.88 of the survey's widths high with an acceptable fit, while DES Y3 stays at 2.18/2.23 on a re-scanned fit. These readings rest on post hoc amendments: the lane's amended label is GAP STAYS and its frozen labels are NO VERDICT. With fixed-amplitude reference runs spliced in at low k, an interim reading on one full-size pair puts DES Y3 at 1.85/1.83 and both surveys inside two widths; that lane's frozen label is NO VERDICT, and its amended label, graded after version 1.1 on two full-size pairs, reads AMPLITUDE CONSISTENT, is post hoc and awaits an independent re-grade. The two failing dwarf classes do not survive the record's own systematics budget, and a gas-history reading does not explain them. The form's data rows are computed without the joint between its two states, and several of them fail or have no verdict. The form is not complete.

Version 1.2 adds the lanes that landed after version 1.1 (Section 8). The Lyman-α forest row, computed for the first time at the forest's own redshifts, FAILS on both footings ($\Delta\chi^2$ 9916/9009 against a line of 4); an independent blind build of the same row also FAILS, no a_0 that changes with redshift passes, and one history of the settled radius leaves no window between the forest and galaxy lensing, though a blind re-grade of that lensing gate returns no verdict. At the level of the transmitted flux, two independent builds of the forest row both return NO VERDICT as frozen. The flux is less sensitive than the matter-level projection because the mass the law moves sits in saturated gas; the blind build's reported reading still misses the line by about a hundred times, and the first build's much smaller miss rests on a pressure-smoothing scale about four times the published one. No lane finds the joint between the falling and the standing clock, derives κ or the share 5.41, or identifies the medium, and no lane turns a failing gate into a pass; the lanes whose labels read as passes are listed as awaiting an independent re-grade or together with their blind re-grade.

Version 1.3 adds the lanes that landed after version 1.2 (Section 9). It states a candidate for the sixth rule, the joint between the falling and the standing clock, and it is NOT ADOPTED: the law's layout holds inside a settled radius in systems slower than a crossing speed $v^*(z) = v^*(0)\,H_0/H(z)$, with $v^*(0) = a_0/(10(1{+}5.41)H_0)$, 668.5/807.9 km s^{-1}; faster systems keep their share as cold dust in the reference layout. The speed follows from rules 2, 4 and 5 and the virial relation with no new constant, and it carries both fitted constants. Graded row by row against everything tested, the completed form fails as frozen on one line, the coverage of massive discs at $z \approx 2$ where the data have no power; most of its passes are trivial or equal to the law's; form B's failing rows stay; and no hit has had an independent blind re-grade that confirmed it. At the level of the transmitted flux the forest row now has graded labels: the law's full layout FAILS; the record's gate fits no gas parameter, and refitting them cuts the full layout's number about twelve-fold without passing it; the candidate's split holds at two settled radii as frozen, fragile on three of four. A blind, independent build of the same split reads NO VERDICT as frozen on every split case, because one of its twins does not bite, with its numbers lower still and under the gate and its control failing: it neither confirms the hold as a label nor contradicts it, so the hold stays a hit, not a confirmed hold. Checks of the data's own assumptions leave the Milky Way's hot bin, the resolved discs at $z \approx 2$ and the Euclid lens gate undecided. Four predictions of the candidate are sealed, with kill lines and a bar on the data that may test them; the curves on disk lean against one of them and decide none. We list what is open, the measurement that would decide each item, and our own wrong estimates.

What changed since version 1.2. Version 1.2 (DOI 10.5281/zenodo.24182757) is unchanged in every sentence that is still true; version 1.3 adds Section 9 and dated additions marked "Version 1.3" in Sections 4 to 7. Section 9 states the candidate sixth rule with the status of each of its numbers (Section 9.1), a compact scorecard of the completed form against 45 rows (Section 9.2), the forest at flux level (Section 9.3), the checks of the data teams' assumptions (Section 9.4) and four sealed predictions (Section 9.5). The candidate is NOT ADOPTED; adopting it is the owner's decision. $\kappa = \frac{1}{2}$ and 5.41 are fitted, the theory is not closed, and a hit that no blind re-grade has confirmed is called a hit.

1 Two forms

Cold dark energy is this record's name for a gravitating field component with $w \approx 0$ at the cosmic amount; it is not a particle. Two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alt 1.1312×10^{-10} m s^{-2}, are carried throughout and never pooled; pairs of numbers are canonical/alt unless a row says otherwise.

Form A, the law alone, is what PAPER70 states. It passes rotation curves and the Solar System. It fails the CMB (which needs a cold component of 5.41 times the baryons), growth, X-COP by +0.4675/+0.4345 dex, the Bullet ($Z = -4.55$, the mass sits on the gas) and Coma's ultra-diffuse galaxies (+1.12 dex with a static external field, +1.41 with a moving host).

Form B, the law with cold dark energy counted once, is the single-counting reading that lanes PD1 and PD2 graded with one fixed share per halo. As graded there it is FAIL/FAIL: it passes its population budget, rotation curves (rms 0.0563/0.0558 against a line of 0.0660) and the carried cosmology, and fails KiDS's pooled outer points (worst −0.635 dex), the groups and the Bullet's amount; X-COP passes only in a reported bracket (+0.019/+0.010).

On 2026-11-09 the owner chose B as the form the record closes on, with both forms scored side by side from the record. The choice is of a form, not of a pass: no label was changed by it. It reopens the cold sector retired on 2026-11-01 for this one reading. Section 3 is the regrade that followed.

2 Form B in the adopted action

The action of PAPER70 has certified field equations (lane QX17). A note written after the decision, and checked independently for this paper, reads form B out of them for a static clock.

- **The law's halo is energy density.** The khronon sector's energy density contains $\Pi = D{\cdot}P$, and the static weak-field 00 equation is Poisson's with source $\rho_b + D{\cdot}P$. Light and matter both see it.
- **One conserved charge.** The khronon equation is a conserved current whose charge is, to leading order, $D{\cdot}P + \mu_K^2(Q-1)/K$. With $P = 0$ it is cold dust in the expanding background; around a static source it is the law's halo plus the static dust term $-\mu_K^2\Phi/K$, which is zero at $\mu_K = 0$. Nothing is counted twice.

- **The static state is in force balance.** With $J(g) = (a_0^2/K)[u - \ln(1+u)]$, $u = g/a_0$, the law sector's stress satisfies $\partial_j \Pi_{ij} = (D \cdot P)\, g_i$. In the deep field $J = g^2/2K$, which is the pressure of an isothermal halo with $\sigma^2 = \frac{1}{2}\sqrt{GM_b a_0}$: the relation PD1 had to assume. The identity holds for any $J(g)$ with a curl-free field, so it shows that the action's static state balances itself; it is not evidence for this J.
- **Force balance is not stability; the stability was then computed.** With the metric frozen (the decoupling limit) and in a planar background, lane TPA1 found the action's law-on phase gradient-unstable, with no ghost, wherever the clock's own dust density is below a critical value; the e-folding times on its grid are 5.0 Myr to 0.62 Gyr. The note on form B read the form's static state, which has almost no dust of its own, as lying in that range. Lane SCS1 then coupled gravity around a point mass and labels the state STABLE on both footings: two independent routes find no growing mode at any wavenumber, and disturbances oscillate at 0.08 to 12 times the local free-fall rate. The clock's gradient coefficient is its own charge density, $n_d + D \cdot P$. In TPA1's planar background $D \cdot P = p'$, which is negative; around a point mass $D \cdot P = p' + 2p/r$ is the phantom density, which is positive. SCS1 reproduces TPA1's numbers in the planar background, and the growth returns when that background is put back into the coupled system. TPA1's other findings stand, and the note's sentence has been corrected.
- **What that does not settle.** The result is linear, and for point masses only. For flattened baryons the law's phantom density can be negative in places, and there the same closed form gives growth; SCS1 did not analyse discs or extended spherical distributions. Version 1.1: lane SCS2 analysed discs. Isolated discs (razor-thin exponential and Kuzmin at four masses, a thick disc, a disc with a bulge) are STABLE on both footings: $D \cdot P$ is positive in every cell on both grids and no mode grows on either route. SCS2's frequency known answer fails with its frozen fit and passes under its fourth amendment, which is post hoc and followed a graded run. With a uniform external field along the axis or in the plane the label is NO VERDICT on both footings: both routes find unstable regions, and the magnitudes are not converged or rest on one grid. Lane SCS3 took the external-field case alone and also returns NO VERDICT on both footings; it says that the sign is not in doubt and that STABLE is excluded there. It also says that SCS2's label for isolated discs does not cover the centre, and that whether the static clock is stable at a disc's own centre is open. Lane DCN1 then took the centre. Every reading it froze before its outcome was seen is NO VERDICT, each on a known-answer gate; under the last of its six post hoc amendments, two of them written after the outcome was seen, it reads ARTEFACT on both footings, with the isolated centre stable. Whatever the reading, every one of its 24 resolved solves has $D \cdot P$ positive in every cell and no growing rate. Extended spherical distributions remain unanalysed. At the dust constant the CMB needs, the static state is Jeans-unstable at the local free-fall rate (e-folding times of 17 Myr to 3.9 Gyr): that is the cosmological setting and not the form's galaxy setting. The lane's three amendments are post hoc and disclosed; one mutation arm does not bite, being a field redefinition, as the lane said before any run.

What the action does not give is stated in Section 5, items 1 to 4.

3 The regrade

Each lane froze its criteria in a commit before any graded number was computed (KO2 ran a model-only design toy before its freeze and says so), and has known answers that must pass and mutation twins that must fail. KO2, RB1, RB2, CO1, CO2, HE1, TPA1 and ONE1 were each re-run by a session other than the one that wrote it, and their labels reproduced; the re-runs are listed in `closure_2026_11_09/RERUNS_2026-11-09.md` and `fable52_push/README.md`. SCS1 (Section 2) landed last: its lead session re-ran both of its routes and recompiled its Lean file (13 theorems); the first route reproduces with only timing lines differing, and the second gives every label identical and every eigenvalue equal to better than one part in a million. The lanes relabel nothing. The lanes added in version 1.1 are in Sections 2, 5 and 7, and Section 7 says which of them were re-run; those added in version 1.2 are in Section 8, which says the same for them.

3.1 Groups, clusters and the Bullet (RB1): NOT FALSIFIED (ONE-SIDED) on both footings

In form B a hot system holds $R = 5.41$ times the baryons that fell into it, and the component stays when gas is expelled or stripped. The retained fraction $f_{\rm ret} = R\, M_{b,\rm obs}/(M_{\rm dyn} - M_{b,\rm obs})$ can therefore not exceed 1 at a system's edge. This gate can fail from above only; the two-sided amount gate stays FAIL as PD1 graded it.

- **Groups.** The 20 hydrostatic groups read $f_{\rm ret} = 0.597$ [0.556, 0.637] at R_{500}, none above 1. Form B reads the missing 40 % as gas outside R_{500}. Nothing on disk measures it.
- **Clusters sit at the ceiling.** X-COP at its last radius reads 1.095 [0.885, 1.259], with 4 of 7 clusters above 1. At R_{500} on the forward-fitted mass the class reads 1.057 [1.046, 1.072], above the ceiling beyond its errors; this row is a reported bracket, and graded there the gate would fail. On the NFW-fit hydrostatic mass the same class reads 0.959. The verdict for the deepest wells turns on about 10 % in the hydrostatic mass.
- **Trend.** $f_{\rm ret}$ rises with depth: slope +0.160 [+0.080, +0.225] in $\log M_{500}$ from groups to X-COP.
- **Radial shape.** One collisionless component with the record's own profile, anchored at each cluster's last radius, gives missing-mass fractions 0.72 / 0.63 / 0.55 / 0.47 / 0.35 per r/R_{500} bin against the measured 0.71 / 0.66 / 0.58 / 0.46 / 0.32 (canonical); bin residuals −0.007 / +0.033 / +0.006 / −0.030 / −0.041 dex against a line of 0.10. This label rests on a post hoc amendment; under the frozen known answer the gate is NO VERDICT.
- **The Bullet.** PD1's 2.6σ deficit in the subcluster is the bookkeeping of the stripped gas. With the gas shared in proportion to the stars the amounts read −0.45 / +0.37 (main / sub) and pass; with nothing reassigned they read +0.49 / −2.58 and fail. No measurement on disk fixes the subcluster's gas before the merger.

3.2 Hosted dwarfs (RB2): B FAILS ON THE HOSTED ROWS on both footings

In form B a dwarf keeps the phantom of its own baryons inside a host, cut only by tides. There is no external field effect.

- **What it removes.** Coma's ultra-diffuse galaxies move from +1.120 to +0.326 dex (alt +0.283) and read B CONSISTENT; the interval contains zero only with the systematics budget, and on statistics alone it is [+0.105, +0.547]. The Milky Way's classical dwarfs read B CONSISTENT as a class (+0.281/+0.241), with Draco, Ursa Minor and Sextans high row by row.
- **Two classes fail high.** The Milky Way's ultra-faints sit at +0.634 [+0.433, +0.865] (alt +0.592) and M31's bright dwarfs at +0.273 [+0.085, +0.473] (alt +0.231). The statistic carries measurement scatter only; a full systematics budget was not computed, since that would be a change after the freeze.
- **Rows that cut the other way.** NGC 1052-DF2 and DF4 sit 0.74 and 0.94 dex below B and at Newton on their baryons. Tides reach them only if NGC 1052 holds 17 and about 100 times the mass B gives it inside their projected distances. NGC 147 and IC 1613 sit 0.56 and 0.54 dex below B; the law with its external field leaves them as low. Crater II is the one low row tides can reach.

3.3 Lensing by galaxy type (KO2): NO VERDICT for late and early types on both footings

All 181,477 KiDS lenses were fitted with every catalogued neighbour modelled off-centre (888,795 pool objects), in place of a veto. The statistic is the ratio of the amplitude beyond the inner window to the inner amplitude; the law as a galaxy's own field gives 1 and no extended field gives about 0.

- **Why no verdict.** One of 13 known answers failed (a cross-shear pull of 3.11 among 15 amplitudes). Without it the frozen bracket rule gives NOT SEPARATED for both types. One mutation twin did not bite: the ratio measures the shape of the outer profile, not its level.
- **Early types lie between.** They read 0.487 ± 0.071 in the primary design and 0.739 ± 0.115 with lens and neighbour amplitudes untied: below the law's level in every design and above form B's truncated halo (0.038/0.007 in the primary design).
- **Discs cannot be placed.** They read 0.277 ± 0.183 tied and 1.071 ± 0.199 untied. Form B planted with the measured cross-shear as noise reads 1.030 ± 0.282 untied, so the data cannot tell the law from form B for discs.
- **An earlier lead does not generalise.** Lane KO1, and PAPER71 beside its gate, reported that host-free late types show no outer signal (-0.19 ± 0.39 against 1.34 ± 0.18 for early types). Both labelled it a lead and not a result: seen after the freeze and sharing the veto's edge. Plain stacks of all lenses are flat for both types (late outer-to-inner ratio 0.93 ± 0.13, early 1.09 ± 0.08), though they include the neighbours' signal. With neighbours modelled, the deficit appears only inside the host-free subset (-0.03 ± 0.28), which carries the veto's edge.

A look at the host-free lenses, committed before KO2 had a lensing number, compares form B at PD1's share with KO1's shape ratio. It is post hoc and models no neighbours. At the frozen veto radius form B reads 0.510 against a measured 0.607 ± 0.094, with the law as the lens's own field 4.2σ off on the other side. At the widest

veto, where the edge effect is weakest, form B reads 0.525 against 0.811 ± 0.134, 2.1σ low, with the law 1.4σ off. The note read from it that one share does not fit late and early types together; ONE1's frozen type-difference line passes at every veto, so the graded statistic does not read that. The same note predicted blind that KO2's late-type ratio would be at most about 0.5; the tied design meets that (0.277 ± 0.183) and the untied one does not (1.071 ± 0.199), so it is not confirmed. Lane ONE1 then graded form B on those lenses with every catalogued neighbour carrying the same share: KiDS host-free PASSES/PASSES by its frozen gate at the widest veto, for PD1's share and for that row only. The shape ratio there is 0.811 against a model of 1.003 (-1.14σ; alt 0.809 against 0.847); late types read 0.084 against 0.792 (-1.46σ) and early types 0.882 against 0.999. At the frozen veto radius, which carries the edge, the model misses (0.607 against 0.924, -2.68σ). The law with no end gives 0.98 at the widest veto, also inside 2σ, so the pass does not separate form B from the law, and the late-type error there (0.42) is too large to test discs. The look's 0.510 to 0.525 was mainly a matter of calibration: with the inner calibration fitted, the model reads 0.86/0.73 at the widest veto without neighbours and 1.00/0.85 with them. Re-run by its lead session: the KiDS script's label and every graded number reproduce, with only timing lines and one status flag differing; its Lean file recompiles (10 theorems) and its twin fails.

3.4 The last way of keeping the component out (HE1): FAILS on both footings

HE1 held the galaxy's share outside a wall as an envelope in pressure balance with the law's field. The envelope exists, is stable and fills 0.2 % of space or less. It fails KiDS's late-type stacks: none of its three stated shares, at any outer radius, passes its first four gates together. Post hoc and with low power, at the widest veto the record's share fits those stacks (p 0.73/0.83), as does the law. Its fifth gate fails as frozen on two clusters near R_{500} only, and the label does not rest on it. Re-run by its lead session: both labels and every number in `he1.json` reproduce; only timing lines differ. Lane TPA1 wrote out and certified the covariant equations of the two-phase action (EQUATIONS COMPLETE; health UNHEALTHY): its switch wall and its dust phase are healthy, and its law-on phase has the instability described in Section 2, which SCS1 finds does not hold around a point mass. With PAPER71's closed switch class, no reading tried that keeps the component out of galaxies passes the record's gates.

3.5 The cosmology rows, computed (CO1): ROWS HOLD on the amended reading, with an inherited halo list

The decision listed form B's cosmology rows as carried from the cold reference. A further lane computed them. Form B's linear content passes the primary CMB on raw Planck PR4 and ACT DR6 spectra ($-65.21/-65.07$ against a line of 9; cold to baryon 5.410/5.412, fitted). That row replays a fit stored by an earlier lane, and its yardstick is the likelihood package's own test point, not a framework quantity. PR4 lensing reads $0.9617 \pm 0.0231/0.9564 \pm 0.0230$ of the predicted amplitude ($-1.66/-1.89\sigma$ against a 2σ line; shape χ^2 8.56 over 9 bandpowers), and DESI DR1 $f\sigma_8$ passes (largest pull 1.22/1.23). A cold component of the fitted amount gives the cold-component spectra, so these three rows holding is not independent evidence for the framework. For the layout frozen in the criteria (galaxies: the Lagrangian share as the law's phantom; groups and clusters: the share inside R_{200c}, part of it kept in the reference profile) the matter power changes by 0.6/1.0 % at $k = 1$ and 5.1/8.8 % at $k = 3\,h\,\mathrm{Mpc}^{-1}$, and raw KiDS-1000 cannot see it on the 104 points the reference run supports.

- **Two readings of the label.** One mutation arm did not bite as frozen; the label rests on an amendment committed before any graded number. Read as frozen it is NO VERDICT.
- **The halo list is inherited:** a mass function, a bias and a concentration relation calibrated on simulations. The redistribution rows are externally calibrated and say so.
- **The scope decides it.** If a cluster held the whole dust of its turnaround sphere, the power at $k = 1$ would be 2.9 times higher and raw KiDS would reject it. The form's cosmology needs the cluster share tied to the baryons present (5.41 times them) and counted inside R_{200c}; RB1 grades the same bookkeeping at R_{500} or the last measured radius.
- **CO1 left the amplitude open.** The fit's σ_8 is 0.8405/0.8429, with no low-multipole data to hold it. Raw KiDS fits the record's reference run at 0.811 with $p = 0.014$ and form B at its own amplitude with $p = 0.0015/0.0009$ (a bracket with constant-factor scaling, which understates it). The forest row stays carried. Version 1.2: the forest row has since been computed at the forest's own redshifts and FAILS on both footings, and on the record's own halo list the redistribution rows are NO VERDICT as frozen (Section 8.1). CO2, below, pins the amplitude and finds that these absolute p-values carry the reference run's realisation noise.

3.6 The amplitude, pinned (CO2): AMPLITUDE EXCLUDED BY RAW KiDS AT 3 SIGMA as frozen, on both footings; not robust

Lane CO2 refitted the same spectra with the Planck 2018 low-multipole polarisation table added. The optical depth moves from 0.0697/0.0729 to 0.0532 ± 0.0066 on both footings (a width along the degeneracy only, which understates the profile width) and σ_8 from 0.8405/0.8429 to 0.8271; the cold to baryon ratio stays at 5.411/5.412.

- **The CMB side closes.** The pinned fit is adequate ($-63.79/-63.75$ against a line of 9). PR4 lensing reads $0.9928 \pm 0.0239/0.9929 \pm 0.0239$ of the prediction (-0.30σ on both footings, from $-1.66/-1.89$ unpinned), and DESI DR1 $f\sigma_8$ passes (χ^2 1.95 over four bins).
- **Weak lensing is where it fails.** For form B's spectrum raw KiDS-1000 is best at $\sigma_8 = 0.7358 \pm 0.0162/0.7341 \pm 0.0161$, and the pinned value sits 5.62/5.78 widths above it; with one 0.06 eV neutrino (a bracket at the pinned parameters, not a refit) σ_8 is 0.8152 and the gap is 4.89/5.04 widths. The widths are KiDS's alone. Across every variant below the gap is 5.1 to 5.8 widths, and the lane calls it its robust statement.
- **The label is at its line.** The graded row is an absolute fit. Raw KiDS at the pinned amplitude, on the record's gravity-only run moved by the run's own response, reads χ^2 157.89/159.35 on 104 points, $p = 0.0026/0.0021$, just below the frozen 3σ line of 0.0027. With a constant-factor move (a bracket) the same row reads $p = 0.0044/0.0037$, tension and not exclusion; the reference run alone at the same amplitude reads χ^2 156.88/156.85; with the neutrino bracket $p = 0.0090/0.0077$. Unpinned, the same row reads $p = 0.0004/0.0002$, so the pin moves the form towards the lensing data and not far enough. DES Y3 through the same path reads $p = 0.050/0.051$ (reported, not graded).
- **The exclusion does not survive a check made after grading.** The reference run is a single box of $100\,\mathrm{Mpc}/h$ whose large-scale power is a high draw: its starting spectrum is 1.48 times its own linear spectrum in its third bin. A difference of two models on the same run cancels this; an absolute fit does not. With the realisation taken out in four ways (the run's own starting ratio with two tapers, and two $200\,\mathrm{Mpc}/h$ runs below $k = 0.5\,h\,\mathrm{Mpc}^{-1}$), p is 0.003 to 0.013: tension each time. The check is post hoc, is reported and changes no label; the lane says it would not quote the exclusion without it. CO1's absolute KiDS p-values carry the same noise (its reference row, $p = 0.014$, reads 0.021 to 0.049 corrected); CO1's label rests on a difference and is not affected.
- **What the label covers.** Form B as stated, a gravity-only run with nothing ejecting gas beyond R_{200c}, and the inherited halo list; the fit keeps massless neutrinos and the shear row uses 104 of the 225 published KiDS points. The lane's criteria say, before any number, that a fail here is the weak-lensing amplitude tension that any cold-component fit to the CMB carries: it is not peculiar to form B, and form B does not relieve it. The two lines (0.05 and 3σ) are conventional but were chosen with CO1's numbers in view, which the lane states as a limit. Its one amendment, a known-answer tolerance, is post hoc and was committed before any graded number; the known answer passes the frozen tolerance too.

4 Form B after the regrade

Gate	Reading for form B	Lane
Rotation curves	pass, rms 0.0563/0.0558 (line 0.0660)	PD1
Solar System	pass, carried	PD1
CMB peaks, CMB lensing, growth rate	hold when computed; NO VERDICT with one mutation arm as frozen; redistribution rows on an inherited halo list; on the record's own halo list NO VERDICT as frozen, rows hold under a post hoc amendment that awaits an independent re-grade (version 1.2)	F1, CO1, HLF1

Gate	Reading for form B	Lane
Forest (version 1.2)	FAILS at the forest's own redshifts, $\Delta\chi^2$ 9916/9009 against a line of 4; an independent blind build FAILS (8142/7686); no $a_0(z)$ passes; with the law's layout only inside a settled radius FAILS, FRAGILE; one settled-radius history WINDOW EMPTY; at the level of the transmitted flux NO VERDICT as frozen on two builds, the blind build's reported reading FAILS ($\Delta\chi^2$ 363/425), and the first build's much larger softening rests on a smoothing scale about four times the published one; version 1.3: at flux level the full layout FAILS ($\Delta\chi^2$ 269/288), FAILS AFTER REFIT of the gas template (23.0/23.9) under a post hoc amendment, and the candidate sixth rule's split HOLDS at $f = 2$ and $f = 3$ as frozen, FRAGILE on three of the four, a hit; its blind re-grade reads NO VERDICT on every split case (a twin does not bite), with every split case under the gate and its control failing, so the hold is neither confirmed as a label nor contradicted	FRZ1, FRZ1R, FRZ2, FZS1, SRH1, FRZ3, FRZ3R, FRZ4, FRZ5, FRZ6, FRZ6R
Amplitude of fluctuations, pinned	CMB lensing (-0.30σ) and growth pass; raw KiDS prefers σ_8 0.736 against 0.827, 5.1 to 5.8 of its widths; absolute fit excluded at 3σ as frozen ($p = 0.0026/0.0021$), tension once the reference run's realisation is corrected	CO2
Weak-lensing amplitude on current releases (versions 1.1, 1.2)	KiDS-Legacy and DES Y3: NO VERDICT as frozen, GAP STAYS post hoc (KL1); with fixed-amplitude reference runs NO VERDICT as frozen, AMPLITUDE CONSISTENT under a post hoc amendment that awaits an independent re-grade (BX1); on DESI DR1 lenses NO VERDICT as frozen, AMPLITUDE CONSISTENT, FRAGILE post hoc, and the same word, FRAGILE, from a blind re-grade that says it is not fully blind, with DES Y3 and KiDS-1000 plus DES Y3 in TENSION (LWB1, LWB1R); a consistency check, not a pass	KL1, BX1, LWB1, LWB1R
Clusters, amount	not falsified (one-sided); at the ceiling; two-sided gate FAIL; what the non-baryonic mass is: NO CANDIDATE (version 1.2)	RB1, PD1, CLS2
Clusters, radial shape	consistent under a post hoc amendment; NO VERDICT as frozen	RB1
Groups	not falsified (one-sided), $f_{\rm ret}$ 0.597; two-sided gate FAIL; the census to turnaround NOT EVALUABLE on the lensing on disk and on DESI groups	RB1, PD1, GO1, GO2
Bullet	consistent only if the subcluster's stripped gas is counted	RB1
Coma UDGs	consistent on the systematics budget	RB2
Milky Way classical dwarfs	consistent as a class	RB2
Ultra-faints, M31 bright dwarfs	FAIL high	RB2
DF2, DF4, NGC 147, IC 1613	against the form; tides cannot reach them	RB2
KiDS lensing by type	NO VERDICT; early types between the law and the form	KO2
KiDS host-free lenses	passes at the widest veto with neighbours carrying the share; does not separate the form from the law	ONE1
Outer cluster shear	FAIL, about 3σ with systematics	ZN4, FR1
Static state reached by collapse	BREAKS DOWN (version 1.1): the falling clock folds at the centre and the equations give no answer beyond	COL1
Stability of the static state	linearly stable around a point mass and, away from the centre, around isolated discs with gravity coupled; a disc in an external field NO VERDICT with STABLE excluded; the earlier instability belonged to a planar background; its nonlinear outcome NO VERDICT (version 1.2)	SCS1, SCS2, SCS3, TPA1, EFN1
One dust term for the CMB, clusters and galaxies (version 1.1)	NO VALUE for a quadratic and for an exponential term; keyed on the expansion BREAKS AT TURNAROUND; keyed on the acceleration BREAKS AT A1; gated on the halo density NO VALUE as committed, which an independent audit finds not established; computed blind, CMB-ONLY: galaxies pass only with the gate shut, and X-COP passes for no gate (version 1.2)	MK1, MK2, TK1, AKD1, PIK1, PIK2

Gate	Reading for form B	Lane
The joint (version 1.2)	not found: every lane that grades a carrier or a settling route fails, has no verdict, or reads otherwise only under a post hoc amendment that awaits an independent re-grade (Section 8.2); version 1.3: a candidate, the crossing-speed joint, is stated and NOT ADOPTED; it FAILS ON COVERAGE ONLY as frozen in both builds (Section 9)	EDG1, SET1, VSH1, VSJ1, AKS1, CLOSE2, JNT2, JNT2R, ...
Milky Way, DESI DR2 and the literature (version 1.2)	the line FAILS HOT, FRAGILE against the halo stars directly, where every smooth model, DESI's NFW included, fails in the same bin (RAW1); the enclosed-mass profile passes and the extrapolated M_{200c} fails (DW1); the line passes the shape-measuring curves (NMB1), and at three radii the canonical line with the field FAILS LOW, FRAGILE (NIM2); the outer masses PASS, FRAGILE (PLT1); hot gas does not supply the rotation curve (MWG1, HG2), and the stellar mass it needs is above every published central value (MWI1, RCF1, STM1); NMB1, PLT1, MWG1, MWJ1 and MWI1 reproduced by an independent re-grade (RG1); every field-case reading is CONVENTION-DEPENDENT (EFC1); the owner adopted the actual-field reading after EFC1, and row 32's mass kill is retired; version 1.3: the hot bin's fail depends on an assumption the data cannot test (MWT1), a structure test has no power (MWT2), and with the measured RR Lyrae slope the hot bin passes and the 40 to 55 kpc bin turns cold for every smooth model (RRL1)	MWT1, MWT2, RRL1, RAW1, HOT1, LMC1, DW1, NMB1, NIM2, PLT1, SAT1, MWG1, HG2, MWI1, RCF1, STM1, MWJ1, RG1, EFC1
Euclid strong lenses (version 1.2)	the law on the stars alone: NO VERDICT on the full sample, at its ceiling; WIDE LENSES BREAK THE CEILING, confirmed by a blind re-grade; source redshifts and dispersions NOT EVALUABLE; wide lenses sit in richer environments, confirmed by a blind re-grade (EUC6, EUC6R); the wide lenses' extra mass is extended, its slope not constrained (EUC8); population source gate NO VERDICT, confirmed by a blind re-grade (EUC13, EUC13R); a cored gas body excluded, a cusp excluded only post hoc (ESZ1, EXR1); version 1.3: the source-redshift gate graded for the first time, NO VERDICT with the centre on the failing side (EUC16)	EUC16, EUC1, EUC4, EUC4R, EUC6, EUC6R, EUC8, EUC2, EUC3, EUC9, EUC10, EUC5, EUC5R, EUC12, EUC13, EUC13R, ESZ1, EXR1
Flat a_0 at $z = 2.33$ (version 1.2)	NO VERDICT as frozen; FLAT HOLDS / EXPANSION-MODEL DEPENDENT under a post hoc amendment; does not separate the framework from ΛCDM; a blind re-grade reads the same (FLAT FAILS at Ω_m 0.25 and 0.35)	AZ2, AZ2R

5 What stays open

1. **The collapse.** For a static clock the action's equations give the law's shape and hold it. No solution takes a free-falling clock, dust falling in with the baryons, to that state. This is the spiral wall of the 2026-11-01 ledger. Version 1.1: lane COL1 computed it and labels it BREAKS DOWN on both footings. A clock that starts as cold dust in the Hubble flow at $z = 10$ around a baryon point mass folds at the centre within 0.063 to 0.154 Gyr, at the three dust constants its brief asked for and at all four masses, and the equations give no answer beyond that moment; one of its known answers fails, so its law-off rows and one bracket are NO VERDICT. Lane FLD1 (DOES NOT on both footings) finds that the folded region today holds every SPARC galaxy's last measured radius (175 of 175) and every galaxy of the high-redshift sample on disk at its stated extent (100 of 100), and stops 1.3 to 5.4 times short of the KiDS windows and 1.1 to 4.2 times short of the form's own cap (canonical). Lane REL1 tried the law on the clock's acceleration relative to the baryons: NOT WELL-DEFINED, since matter's equation of motion becomes fourth order. Lane LFS1 asked whether the clock's leafwise flux can supply the halo: ILL-POSED, and the answer is no; the flux adds no charge. Version 1.2: no route tried reaches the static state. Settling by relaxation towards the law's pressure STRUCTURE FAILS AT A1 (SET1); no local dissipation of the clock closes the joint (AKS1, a post hoc part); a viscous shock settles the clock only under a post hoc amendment (VSH1, which awaits an independent re-grade), and that settled state is not the joint (VSJ1, also post hoc); an engineered kinetic function FOLDS (CLOSE-1);

and an edge layer cannot hold the charge (EDG1, FAILS AT E3). Version 1.3: the candidate sixth rule of Section 9.1 places the two layouts by a crossing speed; it does not say how a falling clock becomes a standing one, and nothing derives a sharp switch or its width.

2. **Stability beyond a point mass.** With gravity coupled the static state is linearly stable around a point mass. Version 1.1: isolated discs are stable too away from their centres (SCS2). A disc in a uniform external field, where the law's phantom density is negative in places, has no verdict (SCS2, SCS3), and SCS3 says that STABLE is excluded. A disc's own centre is open, and the nonlinear outcome has not been computed. Version 1.2: lane EFN1 took the nonlinear outcome in an external field and reads NO VERDICT on both footings in every reading: its direct route stops at an artefact of its own boundary, and its weakly nonlinear route is not converged.
3. **One dust term.** With the quadratic $K(Q)$ the background has $w = w_0/(w_0 + a^3)$ with $w_0 = I_0/4\mu_K^2$. Our arithmetic, not a lane, with H_0 70 and Ω_c 0.26: $w_0 = 1.063 \times 10^{-8}/\mu_K^2$ (μ_K per Mpc), so at $\mu_K = 3.71$ per Mpc the component still has $w = \frac{1}{2}$ at $z = 1089$. Dust by recombination needs μ_K well above that; the galaxy-lensing bound is 0.1. Version 1.1: lane MK1 labels the question NO VALUE on both footings. The CMB needs μ_K of at least 198 per Mpc for the component to be cold by $z = 3400$ and 2.51×10^4 for its lensing shape, far above anything the galaxy rows allow. The 198 per Mpc line is a background line with the polarization off, and an independent audit of 2026-11-12 (`reviews/codex_corrections_audit_2026_11_12/`, C1) finds that a later correction cannot touch it. The 2.51×10^4 line was imported from a calculation with the polarization response switched off; with it on, and while the clock's gradient stays well below a_0, the cold dark energy's self-gravity factor is $\mu_K^2/(\mu_K^2 + k^2)$, not $1 - k^2/\mu_K^2$. No replacement lensing likelihood has been run; near 2.5×10^4 per Mpc the two factors agree to first order, and the auditor's own estimate, which is not a likelihood, puts the corrected line between about 1.5×10^4 and 2.5×10^4 per Mpc. MK1's NO VALUE does not rest on that line: the 198 line alone sits a factor 198/259 above KiDS's bound, and X-COP allows no value. MK1 also corrects the bound quoted above: 0.1 is the first failing grid value, and the bound lies between 0.03 and 0.1. Lane MK2 returns NO VALUE for an exponential dust term (as frozen NO VALUE/NO VERDICT). A stiffness keyed on the clock's expansion BREAKS AT TURNAROUND (TK1), and one that falls with the clock's acceleration BREAKS AT A1 (AKD1); Section 7 gives both. Version 1.2: a dust term gated on the law's own halo density (PIK1) reads NO VALUE on both footings; its file says the verdict is carried by an analytic derivation evaluated on the record's published numbers, not by a fresh numeric pipeline, and that a re-run is owed. An independent audit of 2026-11-12 (C2) finds that its criteria, result and scripts were added in one commit, that one committed script does not compile and that no output was ever made; that its step showing the gate invisible at linear order holds only for gates flat at the origin; and so that its NO VALUE is not established and its ordering of clusters against galaxies is a strong heuristic, not a graded row. Lane PIK2 then computed the gated term's rows blind: CMB-ONLY on both footings at $\mu_K = 2.512 \times 10^4$ per Mpc, the same at 198.25 and, reported, at 8.89×10^6 per Mpc; no gate and threshold opens a window. In its own words, galaxies are passed only by shutting the gate everywhere the law is measured; X-COP passes for no gate, at no threshold and at no μ_K (best rms 0.39 against a line of 0.10); where a gate gives the groups their dust, the gated term is unhealthy there by 10^{13} to 10^{18}, at fixed metric; and the gate is visible at linear order about FRW if and only if $S'(0)$ or $S''(0)$ is non-zero, the gates with $S''(0) > 0$ failing linear health at every threshold. Its comparison agrees with the audit on that visibility and with the direction of PIK1's cluster ordering, not with PIK1's mechanism or numbers: what binds is the KiDS outskirts, where the groups' threshold sits 1.15 to 1.5 dex above what the galaxy rows allow. It lists as not computed a CMB-lensing likelihood, the coupled-gravity spectrum, SPARC and KiDS through MK1's own graders, the full constrained instability and the a_0-saturation regime. A re-run by the lead from a copy reproduces it; that is not an independent re-grade.
4. **Clusters are not a static-clock state.** Closing X-COP with the static dust term needs μ_K of 1.78/1.74 per Mpc, 17 to 18 times the galaxy bound. No other state has been solved. Version 1.1: MK1 corrects the sentence before: 1.78/1.74 is only where the mean residual is zero, the shape passes at no μ_K, so the dust does not close X-COP at any value, and '17 to 18 times' should read '18 to 60 times'; done as the action's own static equation the mean-zero value is 1.53/1.46. Lane CLS1 searched the action for a cluster state. As frozen its label is STATE EXISTS-PARTIAL, with 0 of 7 clusters holding a state; at the form's own dust constant (198.25 per Mpc and above) it is NO STATE on both footings, proven on all seven. Version 1.2: lane CLS2 asked what the clusters' non-baryonic mass is under form B and reads NO CANDIDATE on both footings: three candidates are killed by proof and the fourth, gas that is under-counted, fails on a budget.
5. **Groups** are untested from below: form B needs 40 % of their baryons to lie outside R_{500}. Version 1.1: lane GO1 tried the census in group outskirts and is NOT EVALUABLE: the lensing on disk does not reach the turnaround radius. Version 1.2: lane GO2 repeated it on isolated DESI groups and is again NOT EVALUABLE

on both footings; where the turnaround radius can be located at all, the error on $f_{\rm ret}$ is 0.9 to 1.8 against a gate of 0.20.

6. **The richest clusters** sit at or above the form's ceiling, depending on the hydrostatic mass used. Version 1.1: lane XCV1 regraded X-COP with a mass that does not assume hydrostatic equilibrium: NO VERDICT as frozen, since a known answer fails; under its post hoc amendment the class is AT THE CEILING with the central value above it.
7. **The Bullet** passes only with gas credited to the subcluster that no measurement fixes.
8. **Ultra-faints and M31's bright dwarfs** sit above the form. Version 1.1: with the record's systematics budget and the dispersion upper limits counted, neither class's interval excludes zero (DG1); the medians stay high. A blind regrade with a full budget (DSB1) reads NO POWER for every class under its amended reading. Version 1.2: three readings that would put the ultra-faints low (stripping, an external field registered at formation, falling in at the host's field) each FAIL THE PAIR with X-COP on both footings (CRACK-1): each predicts them low, and they sit at +0.592/+0.496.
9. **DF2 and DF4** sit below it and need a host 17 to 100 times heavier than the form gives. Version 1.1: lane NH2 set NGC 1052's measured mass against that need: NO VERDICT as frozen, since two known answers fail; under its post hoc reading tides cannot reach DF4 and DF2 is not decided.
10. **KiDS.** On all lenses, early types need more beyond the inner window than PD1's truncated halo and less than the law. On host-free lenses the form passes a gate that the law passes too. Discs are not tested by either. Version 1.2: a blind re-grade of SRH1's lensing gate (SRH1R) finds that the gate passes equally at a settled radius 0.05 of the fold and with the cold reference everywhere, so this lensing test cannot tell one settled radius from another. Version 1.3: that gate sets the lower edge, 1.25, of the candidate's f (Section 9.1), so the lower edge rests on a gate a blind build could not reproduce as a function of f.
11. **The outer cluster shear** stands near 3σ.
12. **Cosmology.** The weak-lensing amplitude is not solved: the CMB pins σ_8 at 0.8271 and raw KiDS prefers 0.736 for the form's spectrum, 5.1 to 5.8 of its widths lower. Form B states neither gas expelled from haloes nor neutrino mass. The forest row is carried, and the redistribution rows rest on an inherited halo list and on the cluster share being counted inside R_{200c}. Version 1.1: lane KL1 took the gap apart (Section 7). On the surveys' current data the gap is 1.71/1.88 widths on KiDS-Legacy, with an acceptable fit, and 2.18/2.23 on DES Y3 (2.89/2.82 on the frozen scan). These are post hoc readings: the amended label is GAP STAYS and the frozen labels are NO VERDICT. Lane BX1's interim reading with fixed-amplitude reference runs puts DES Y3 at 1.85/1.83; its frozen label is NO VERDICT. Version 1.2: the forest row was computed and FAILS (Section 8.1). BX1's amended label, graded on two full-size pairs, is AMPLITUDE CONSISTENT on both footings, with DES Y3 at +1.78/+1.76 widths; it is post hoc and awaits an independent re-grade. On DESI DR1 lenses LWB1 and its blind re-grade read AMPLITUDE CONSISTENT, FRAGILE (Section 8.4). These are consistency readings and close nothing; the item stays open. Version 1.3: at the level of the transmitted flux the law's full layout FAILS the forest row and still FAILS AFTER REFIT of the gas template; the candidate's split holds at $f = 2$ and $f = 3$ as frozen, a hit; its blind re-grade reads NO VERDICT as frozen with its numbers under the gate, so the hold is neither confirmed as a label nor contradicted (Section 9.3). The weak-lensing amplitude has not been computed for the candidate's split.

6 What would decide it

- The spherical collapse of the clock around a baryon mass with the record's own equations was computed in version 1.1 and breaks down (COL1). What would decide the joint now is an equation for what the clock is behind a fold; Section 7 lists what the landed verdicts force on it.
- The stability of the static state for a disc in an external field with converged magnitudes and for an extended baryon distribution, and its nonlinear outcome. Point masses and isolated discs away from their centres are computed and stable.
- The baryons between R_{500} and the turnaround radius around the same groups (thermal SZ, X-ray, stars) with a lensing mass to that radius. Form B needs $f_{\rm ret} = 1$ at turnaround; a census that falls short falsifies it.
- A cluster mass beyond R_{500} that does not assume hydrostatic equilibrium.
- A measured mass for NGC 1052 inside 79 and 165 kpc.
- The dwarf classes regraded with a full systematics budget, frozen before the numbers. Version 1.1: lane DSB1 did this blind. Its frozen reading STOPPED at a known answer and has no label. Its amended reading, with that one known answer replaced after the freeze and before any graded number, is NO POWER for every

graded class on both footings: no class fails, every median stays above zero, and the smallest offset each class could detect (0.59 to 1.23 dex) is above the frozen 0.30 dex line, so the classes cannot decide the form either way. The Milky Way's ultra-faints read +0.572 [−0.043, +1.173] (alt +0.542 [−0.087, +1.127]); reported and not graded, they read as failing high if their nine upper limits are fixed at the limits or dropped. What would decide the dwarf rows now is data with more power, not another regrade.

- Lensing by type with more area. A spectroscopic veto with the redshifts on disk has no power: the late-type error at the widest veto is 0.68/0.62 of the law in KO1 and is forecast at 0.57/0.52 with such a veto, against about 0.47 needed.
- For the weak-lensing amplitude, now that the CMB side is pinned: a refit that carries neutrino mass, in place of the bracket at fixed parameters; the shear row on more than one reference run, and on all of the published points; and a statement by the form of whether gas leaves haloes. Version 1.1: KL1 made the refit and put the measured gas in; it finds the single reference run to be the limiting input, so the deciding step is the shear row on reference runs without realisation noise. Lane BX1 has done this for one full-size pair (Section 7); its second pair was still running when version 1.1 was written and has since finished (Section 8.4). Version 1.2: for the forest row at flux level, a hydrodynamic forest model that sets the gas smoothing scale and a box of at least about 50 Mpc/h (FRZ3R); for the strong lenses, the source redshifts of Euclid's lenses, for which EUC1 froze a gate and which neither EUC2 in DESI's fibres nor EUC3 in Euclid's grism tables could find; for row 32 of the falsifier matrix, an enclosed-mass profile from 100 to 200 kpc with its covariance, which does not yet exist; for the groups, a stack that reaches the turnaround radius, which GO2 did not.
- Version 1.3, for the candidate sixth rule (Section 9): resolved rotation curves of $z \approx 2$ discs reaching beyond about 3 $R_{1/2}$, with measured gas profiles, on both sides of $\log M_b$ 11.3/11.5, fitted by beam-forward modelling with each disc's real point-spread function and point covariance; a blind build of the forest split whose boundary-removed twin bites (FRZ6R's did not), frozen in advance, then a hydrodynamic forest model that sets the smoothing scale; a lensing gate that can see the settled radius; a census and lensing mass to turnaround around groups near $v^*(0)$; the Milky Way's outer star-count slope beyond about 39 kpc with a Sagittarius stream model; and many more certified source redshifts for Euclid's lenses than EUC16's 11.
- Gaia DR4 (2026-12-02), scored as pre-registered.

7 Version 1.1: the theory in six rules, the rows since, and what a completion must satisfy

The lanes in this section landed on 2026-11-10, after version 1.0, except the earlier lanes named in the survey of media. Each lane froze its criteria first; where a frozen reading and a post hoc reading differ, both are given, the frozen one first. The closure session re-ran DG1's main run (not its mutation twin), all three parts of KL1 and AKD1 from copies, and every label and stored number reproduced apart from run times and memory (`closure_2026_11_10/RERUNS_2026-11-10.md`); a re-run reproduces a lane's arithmetic and is not an independent derivation. KL1's second part was also regraded by lane KL1R, with the same labels. The other lanes named here are quoted from their result files and were not re-run for this note; for the lanes under `fable52_push/`, what their lead session checked is recorded in that folder's `README.md`.

7.1 The theory in six rules

1. **Fields.** The metric, one clock field that picks a slicing, and a polarization on its slices. Matter couples to the metric only. The action is lane QX17's action (A), which is PAPER70's.
2. **One conserved charge.** The clock equation is a conserved current. The universe carries 5.41 units of its charge per unit of baryons, a fitted number. It is not derived: in the action it is an integration constant (note R541, which is not a graded lane), and four attempts to derive it are EXCLUDED on both footings: a clock-matter coupling (R541A), a Gauss-Bonnet source (R541C), a baryon-current coupling (R541D) and a coupling to $B - L$ (R541E). A fifth, an early potential on the clock (R541B), reads TRADES: it moves the number into the potential's height and does not fix it.
3. **Falling clock.** Where the clock free-falls with the expansion the charge is cold dust. This is the cold component of cosmology.
4. **Standing clock.** Where the clock stands still around baryons its own equations give the law, with $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted. The halo is the charge itself and gravitates and lenses as mass.

5. **Counted once.** A halo is made of the charge its region brought. A hot system holds 5.41 times the baryons that fell into it. A body that brought no charge has no halo.
6. **The joint.** How a falling clock becomes a standing one is not derived. Version 1.3: a candidate for this rule, the crossing-speed joint, is stated in Section 9.1; it says where each layout holds, it is a rule and not an equation, and it is NOT ADOPTED.

The data rows of this note are computed from rules 3 to 5, and several of them fail or have no verdict. Rule 6 is the missing equation, and the hot branch of rule 5 is a rule and not a solution of the action.

7.2 What the landed verdicts force on the joint

Any completion has to meet all of these. Each line rests on a lane's verdict on both footings; where a sentence is a reported reading and not a label, it says so.

1. **Not a wall at a fixed place.** HE1 FAILS, and PAPER71's switch class is closed.
2. **Not a dust term that depends on the clock's offset alone.** MK1 and MK2 return NO VALUE (Section 5, item 3). The offset is about 10^{-6} both in the background at $z = 3400$ and in a galaxy's well today.
3. **Not the expansion alone.** Lane TK1 keyed the dust stiffness on the clock's expansion: BREAKS AT TURNAROUND. The key keeps the expanding background well posed, with a margin of 1.19 on the Jeans line. A collapsing region passes through small expansion while its clock is still falling; there short waves grow without bound, 2 to 140 Myr before the turnaround, and the region never turns around.
4. **Not a stiffness that falls with the clock's acceleration.** Lane AKD1: BREAKS AT A1. The key is a negative copy of the law's own term. Its weight against the law is $3.9 \times 10^{14}/4.2 \times 10^{14}$ at $z = 3400$ and $4.5 \times 10^{11}/4.9 \times 10^{11}$ at $z = 1100$, so the clock's inertia is negative on every sub-horizon scale there, until $z = 11.6/11.4$. A shape that is flat at zero acceleration has the same sign problem in the outskirts of static wells; that is a reported class bound and not a label.
5. **Not the law on the clock's acceleration relative to the baryons.** REL1: NOT WELL-DEFINED.
6. **It has to survive the fold.** COL1: BREAKS DOWN. Nothing in the action removes the infall energy.
7. **It has to reach further than today's fold.** FLD1: DOES NOT.
8. **It cannot draw the halo from the clock's leafwise flux, and it needs a cluster state the action does not have.** LFS1: ILL-POSED. CLS1: NO STATE at the form's own dust constant.

So no dust term tried that is built from the clock's own first-order kinematics, its offset, its expansion or its acceleration, joins the two states. Of functions of the offset, only a quadratic and an exponential were graded; MK1's statement about any function of the offset alone is a requirement and not a graded result. One kinetic function with a narrow interval of changed compressibility is named in the two assembly notes: its linear and static checks pass there, and whether a forming halo can reach its static state is not computed. It is not a result.

7.3 Media with more structure than one clock

A design note, `closure_2026_11_10/MEDIUM_DESIGN_2026-11-10.md`, which is not a lane, sets the classes of medium the record has tried against four properties the data ask for: conserved and cold; passing through itself; hot and at full share in clusters; locked to the baryons in galaxies. Each class has a failing verdict in its own lane: any local, momentum-conserving, normally self-gravitating medium whose equilibrium is the law (M5: NO-GO, a theorem); two phases (PH1: NO-GO); a superfluid with phonons (QX28: FAILS at G3, G4, G5, G6 and G8); a self-interacting isothermal inner halo (IJ1: FAILS); a light fermion beside the law (HF1: fails its first gate at 5 of 6 masses with 1 undetermined; the lane says this kills the family on its inputs and is not a no-go theorem) and a warm component beside it (WM1: FAIL AT SEEDING, a gate added after its first result, which failed on groups); the tethered medium (TM1: it does not resolve law against clumping, and its scored rows fail or are conditional); and nonlocal, history and global actions (FA3: no candidate survives). Route E, in which cold dark energy clumps everywhere, gravity is Newtonian and the law is emergent, had its cold sector retired by the owner on 2026-11-01.

One class had no lane: a medium in thermal contact with the baryons where their acceleration exceeds a_0. The note works it out as our own arithmetic, not a graded result. The contact needs an opacity $\pi G/a_0$ of $22.4/18.5\,\mathrm{cm^2\,g^{-1}}$. With it the medium's mean free path against baryons at $z = 1100$ is 25.8/31.1 pc and its drag at a relative speed of $30\,\mathrm{km\,s^{-1}}$ is 0.73/0.61 of the expansion rate, so it does not stay decoupled from the plasma; and a cluster's gas is 0.14 to 0.68 optical depths (canonical), so it would be dragged with the Bullet's gas.

Three lanes then took the question as an inverse problem. Lane PAR1 asked which parent action could carry the cold dark energy at all: NO VERDICT on both footings, because two branches stayed open with no member of them written and graded there. Its statement is that no parent action built from one single-valued carrier exists, a theorem that rests on a post hoc amendment, so a working parent would have to carry the cold dark energy on a multi-valued field. It also reproduces the three numbers of the arithmetic above independently. The two open branches were then graded. Lane WV1, wave dust with a split charge: one variant reads STRUCTURE FAILS as frozen; the other is NO VERDICT as frozen and STRUCTURE FAILS under two post hoc amendments. Some of its runs were paused when version 1.1 was written; they finished on 2026-11-11 and add a failing part, a resolution failure at the larger mass, and under the first amendment alone that variant's reading moved from NO VERDICT to STRUCTURE FAILS, which the lane says rests on a resolution failure, not on physics (version 1.2). Lane SMF1, the simplest smooth-frame parent: STRUCTURE FAILS on both footings at its first part. In any region bounded where the frame free-falls, the frame's polarization charge is zero at every time, so the exchange cannot fill the law's halo in net. A post hoc amendment adds that neither failing gate is specific to waves, so a collisionless carrier in the same structure does not escape either one.

7.4 The data rows since version 1.0

- **Weak-lensing amplitude (KL1).** CO2's gap of 5.62/5.78 KiDS-1000 widths came apart into pieces. The largest is the survey's own release: for the form's spectrum KiDS-Legacy prefers $\sigma_8 = 0.7767/0.7748$ where KiDS-1000 preferred 0.7358/0.7341. The largest on our side is the massless neutrinos: a refit of the CMB fit with one 0.06 eV neutrino gives $\sigma_8 = 0.8121/0.8120$. The covariance evaluated at the model and the gas RB1 measured outside R_{500} are smaller pieces. As frozen, the lane's first two parts read NO VERDICT, because a mutation arm cannot bite on DES Y3. Under a post hoc amendment, KiDS-1000 with DES Y3 reads TENSION (KiDS-1000 gap 3.95/4.06 widths) and KiDS-Legacy with DES Y3 reads AMPLITUDE CONSISTENT WITH RAW LENSING ($p = 0.111/0.101$ for Legacy), with Legacy still 2.13/2.30 of its widths below the form, which the lane's frozen list counts against the form. Lane KL1R, a regrade of that part by another session, returns the same two readings; it says its blindness was not clean, since KL1's headline numbers were visible before it began, and that it is not an independent check of the shared machinery.
- **The second pass.** With the reference run's known draw corrected and the cold component responding to the moved gas, the lane reads NO VERDICT as frozen (a mutation arm does not bite, and on alt a known answer fails on DES Y3) and, under three post hoc amendments, GAP STAYS, driven by DES Y3: KiDS-Legacy 1.71/1.88 widths and consistent (2.06/2.23 with the full Gaussian likelihood), DES Y3 2.18/2.23 on the polished scan (2.89/2.82 on the frozen scan), KiDS-1000 3.72/3.85. Held to Legacy's own alignment model under its own prior, Legacy reads 1.57/1.83 and stays consistent. Gas taken from a stellar-mass-selected stack is a bracket and barely moves anything. The lane concludes that the remaining gap is the size of the reference run's realisation noise: two independent $200\,\mathrm{Mpc}/h$ runs spliced in at low k put DES Y3 at 1.31/1.27 against 3.49/3.43 widths. Lane BX1 then replaced those single runs below $k = 0.5\,h\,\mathrm{Mpc}^{-1}$ with a fixed-amplitude run and its inverted-phase twin in a $200\,\mathrm{Mpc}/h$ box. As frozen its label is NO VERDICT on both footings: one known answer fails and one mutation arm does not bite. Under its post hoc amendment the graded label is the mean of two full-size pairs and was still pending when version 1.1 was written, because the second pair had not finished; it landed afterwards and is given in Section 8.4. Its interim reading on the one full-size pair is AMPLITUDE CONSISTENT on both footings: KiDS-Legacy +1.51/+1.72 widths (p 0.26/0.25) and DES Y3 +1.85/+1.83 on the polished scan (p 0.30/0.31), with KiDS-1000 at +3.59/+3.72. The lane puts a single pair's residual noise at about 0.1 width on Legacy and about 0.2 width on DES Y3, so the DES reading is inside that noise of the two-width line; with the full Gaussian likelihood DES Y3 is above 2 on both footings (+2.27/+2.18). The closure session re-ran the lane's known answers, splice, re-grade and grading from the stored spectra, and both readings and every gate number reproduce; the box simulations themselves were not re-run.
- **Hosted dwarfs (DG1): HISTORY DOES NOT CLOSE THEM on both footings.** RB2's two failing classes rested on measurement scatter alone. With the record's joint systematics budget re-derived for the form and the dispersion upper limits counted, both intervals contain zero on both footings: the ultra-faints read +0.550 [−0.090, +1.175] (alt +0.509 [−0.110, +1.141]) and M31's bright dwarfs +0.299 [−0.181, +0.774] (alt +0.258 [−0.237, +0.747]). The medians stay high; this is the standing RB2 gave Coma's ultra-diffuse galaxies and not a pass. A gas-history reading, in which a dwarf keeps the halo of the gas it lost, brings the class medians to −0.045, +0.037 and −0.010 dex and fails the per-galaxy gates: inside the ultra-faints the rank correlation of the residual with the predicted history is +0.06, and a two-number stellar-mass power law fits

better (rms 0.306 against 0.343 dex). The lane is post hoc to RB2's class numbers and says so.

- **The cold component's shape (SHP1).** Its label is THE LAW'S SHAPE ON ROTATION CURVES, LENSING ABOVE; NOT ONE SHAPE (R3b) on both footings. On rotation curves (2700 points in 149 galaxies) everything that is not baryons follows the law's shape in every bin at the standard mass-to-light ratios, the worst bin median within 0.057 dex; at disc ratios of 0.3 or 0.8 that pass fails. KiDS isolated lenses have the same shape at $1.40 \pm 0.03/1.28 \pm 0.03$ times the law with the published stars only, and sit below it with the published hot gas counted as baryons. A separate halo of PD1's 59.5 M_b on top of the law is excluded on every reading.

8 Version 1.2: the verdicts since version 1.1

The lanes in this section landed after version 1.1 was drafted on 2026-11-10; SET1, NBC1 and the five lanes of the TOE battery landed that day and were not in it. Each froze its criteria first unless its row says otherwise. Frozen readings come first and post hoc readings are labelled. A lane whose label reads as a pass, a derivation or a rescue awaits an independent re-grade unless a blind re-grade is named; none turns a gate from fail to pass or closes an open item. A re-run is named only where the lane's file records it, and a re-run reproduces arithmetic; it is not an independent re-grade. An independent audit of 2026-11-12 graded many of these lanes (`reviews/hy4_codex_audit_2026_11_12/`, files A1 to A4: SOUND, SOUND WITH LIMITS, OVERSTATED or WRONG); its grade stands beside the lane's own label. Its plain answers: κ is derived nowhere, and neither is 5.41; the joint is not found and the medium is not identified; no lane in its data slice rescues a failing row. Another session left uncommitted correction notices in several of these lanes' files; an independent audit of those notices is committed at `reviews/codex_corrections_audit_2026_11_12/` (C1 to C4), and this note quotes the audit's own checks, not the uncommitted notices.

8.1 The forest

Version 1.1 carried the forest row from the cold reference, and CO1's redistribution rows used an inherited halo list at $z = 0$. Lane HLF1 rebuilt them on the record's own halo list: NO VERDICT on both footings. Its Part A is NO VERDICT as frozen, since a mutation arm did not bite, and reads ROWS HOLD WITH THE OWN LIST under a post hoc amendment, which awaits an independent re-grade. Its Part B found that the forest gate fails when CO1's $z = 0$ redistribution is held to the forest's redshifts (160.7/1025.6) and passes trivially under the record's fade, so the row turned on how the redistribution evolves to $z \approx 3$.

Lane FRZ1 computed that evolution: FOREST ROW FAILS on both footings, $\Delta\chi^2$ 9916/9009 against the record's gate of 4. Because the law's physical scale is fixed, galaxies' cold dark energy is laid out flatter at the forest's redshifts than in the cold reference, and the power falls: at $z = 3$ the ratio to the cold reference at k 3/5 $h\,\mathrm{Mpc}^{-1}$ is 0.845/0.665 (canonical) and 0.831/0.669 (alt). Every frozen bracket fails too, so the label carries no FRAGILE. Its lead re-ran it from a copy and the outputs reproduce; the file says this is not an independent re-grade. Lane FRZ1R built the row blind, before FRZ1's result was opened: FOREST ROW FAILS (8142.08/7685.51), and its comparison reads AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL. Both rest on the record's inherited mass-function and concentration ingredients.

Lane FRZ2 let a_0 change with redshift: NO FRAMEWORK $a_0(z)$ PASSES, including every power law. The nearest case, $a_0 \propto (1+z)^{0.5}$, reads 252/167, still 40 to 60 times over the gate; the dark-energy-density cases lower a_0 at $z = 3$ and make the deficit worse. Lane A0Z1's $a_0(\chi)$ also FAILS FRZ2's gate. Lane FZS1 laid the law out only inside a settled radius, a rule post hoc to FRZ1's fail: FOREST ROW FAILS, FRAGILE, $\Delta\chi^2$ 38.7/36.7, with the half-radius bracket passing.

Lane SRH1 followed one history of the settled radius, f times the fold radius, against the forest, rotation curves and galaxy lensing: WINDOW EMPTY on both footings. The forest needs $f \leq 0.70$ on the grid (edge 0.722/0.725) and KiDS lensing $f \geq 1.25$ (edge 1.004/1.050). Its post hoc amendment, reported and never graded, opens a window with a redshift tilt, the outcome its criteria named in advance as counting against. A re-run from a copy reproduces its label and is not an independent re-grade. The blind re-grade of its lensing gate, SRH1R, reads NO VERDICT on both footings because two of three mutation twins do not bite; its gate passes at every f in the window and also at $f = 0.05$, so lensing cannot tell one f from another. SRH1R does not relabel SRH1. Lane FINAL4 grades the growth history of the formation reading on the same gate: DERIVED HISTORY EXCLUDED on both footings, with FRZ1's $\Delta\chi^2$, and TILT POST HOC, NOT EXCLUDED. An independent audit of 2026-

11-12 (C1) finds that this history rests on modelling choices its two equations do not derive (the settled radius equal to the fold radius, the settled fraction, the NFW redistribution), and that its table's growth ratio 0.880 at $z = 0.295$ is a 12.0 % suppression, not the 3.6 % its prose states; neither changes its $\Delta\chi^2$ labels.

At flux level. Lane FRZ3 painted rule 5's layout onto simulated fields and computed the transmitted flux. As frozen it reads NO VERDICT on both footings: its painted fields carry only about half of FRZ1's matter-level redistribution, so its matter-level known answer fails. Its flux numbers are reported and grade nothing: $\Delta\chi^2$ 4.71/2.58 at flux level against 4720/4120 at matter level on the same fields, a softening factor of 0.0010/0.0006; its brackets cross the gate, and its pressure-smoothing scale is PROVISIONAL. Lane FRZ3R built the same row blind and independently. As frozen it also reads NO VERDICT on both footings: its matter-level known answer fails at $z = 2.2$, and its graded field is 6.7/7.4 % negative. Its reading T, added by an amendment committed before any flux number was opened, is reported and is not the frozen label: FOREST ROW FAILS AT FLUX LEVEL on both footings, $\Delta\chi^2$ 363/425 against 5035/4377 at matter level, a ratio of 0.072/0.097, with no bracket under the gate (the lowest 35.7/36.6). The two builds agree on the frozen label, on the matter level of the painted fields and on saturation doing the softening; they disagree by about two orders of magnitude on the flux-level number. FRZ3R traces that, post hoc, mostly to FRZ3's pressure-smoothing scale, 80 proper kpc against the published 79 comoving kpc, about four times larger, and to halo placement; the two causes were not run together, and at matched smoothing the builds still differ by about a factor of nine. FRZ3R discloses that the subjects of FRZ3's result commits were visible to it before it began. The lead re-ran both lanes' graders on their committed parts and the outputs reproduce; that is not an independent re-grade.

So, at matter level the forest row FAILS on two builds. At flux level neither build has a graded label. The flux is less sensitive than the matter-level projection, by about an order of magnitude in the blind build, because the mass the law moves sits in saturated gas; no reading at the published smoothing scale brings the row under the gate, and the blind build's reported reading misses it by about a hundred times. What would decide it, as FRZ3R names it, is a hydrodynamic forest model that sets the gas smoothing scale, and a box of at least about $50\,\mathrm{Mpc}/h$. Neither lane closes or rescues the row.

8.2 The joint

Rule 6 (Section 7.1) stays underived, and the medium is not identified. Each lane below took one carrier or one route to settling; the table gives the rest.

- **Settling.** SET1, relaxation towards the law's own pressure: STRUCTURE FAILS AT A1 on both footings. For extended baryons the force is off by exactly $a_0\rho_b$, and cored baryons need a negative cold density in the core. AKS1's Part T, itself post hoc, finds that no local dissipation of the clock settles it into the law's halo by today, with one branch closed on an estimate. VSH1 computed that estimate: NO VERDICT as frozen; SETTLES BY SHOCK under a post hoc amendment written after the first run's outputs were read. It was not re-run by anyone else and awaits an independent re-grade; audit A3 grades it SOUND WITH LIMITS and notes that its charge is 1 to 4 % of the law's halo. VSJ1 then asked whether that settled state is the joint: NO VERDICT as frozen; STANDS BUT MISCOUNTED under a post hoc amendment; VSH1's settled state is not the joint.
- **Holding and sourcing the charge.** EDG1: FAILS AT E3 on both footings; a shell of orbits in all orientations can be held, but infall gives it 19 to 38 times too little angular momentum (on a PROVISIONAL spin of accreted matter). VCC1: the charge can be sourced, but every source that can pay for it is homogeneous, so it cannot make the law's halo: EXCLUDED (A3: SOUND). CLOSE2 graded PAR1's two open branches: NO VERDICT as frozen; CANDIDATE STRUCTURE FAILS under post hoc amendments, with the union CLOSED only in the member sense (A2: SOUND WITH LIMITS).
- **The kinetic function of Section 7.2.** CLOSE-1 computed whether a forming halo reaches its static state with that function: FOLDS on both footings (A2: SOUND WITH LIMITS). An independent audit of 2026-11-12 (C3) confirms two coding errors in it, an extra $1/a$ in the time map and a growth friction of 3 in place of 2. Repaired, its linear gate fails, 0.491 against its line of 0.50, where it read 0.551, and its derived settled radius, never a computed output, is withdrawn. The first central infall times are unchanged, so FOLDS stands, on more kill conditions than before. C3 adds a limit: the frozen fold gate fires for any cold radial infall with no angular momentum, so FOLDS says the function does not escape cold radial collapse, not that a physical halo cannot form. The lane's committed outputs come from its uncorrected scripts.
- **Media.** An elastic solid FAILS AT S0 (ELA1; A2: SOUND); the halo's anisotropic stress cannot pay the lensing (STR1); FINAL-7 reads NO MEDIUM, closed at momentum conservation (A2: SOUND WITH LIM-

ITS); a pinned bootstrap carrier is KILLED (OUT5); a contact between cold dark energy and neutral gas is KILLED, one gate PROVISIONAL (NBC1), the class Section 7.3 worked out as arithmetic only. The TOE battery (TOE0) reads NO TOE in either sense: action (A) has no weakly coupled flat vacuum (QG1) and the law's function is no term of a power-counting-renormalizable action (HZ1); a non-perturbative origin gives the shape only (NPJ1), and a Lifshitz fermion superfluid is KILLED AT G0 (LSF1).

- **The formation reading.** FINAL-1 writes the theory out and says it does not derive the joint; A2 grades it OVERSTATED and finds that the formation reading moves the joint into a boundary condition. FRM1 reads FAILS on both footings on its frozen line, and HOLDS EXCEPT only after voiding one clause that no data could satisfy. An independent audit of 2026-11-12 (C2) finds that its class bands, 14 of 16 (alt 13 of 16), were set after DD1's per-class offsets had been seen, so they are retrospective consistency, not predictions, and that its cluster ceiling holds for baryon loss at a fixed profile with no incoming charge, with X-COP (+0.50) and CLASH (+0.55) above it (+0.32, +0.23; canonical). The reading awaits an independent re-grade.

8.3 External field

EFN1 took the nonlinear outcome of SCS3's external-field instability: NO VERDICT on both footings in every reading, frozen and under its four post hoc amendments. Its deciding route stops at a corner of its own boundary, and its second route is not converged (Section 5, item 2).

8.4 Data rows

- **Weak-lensing amplitude.** BX1's second full-size pair finished after version 1.1. As frozen its label is unchanged, NO VERDICT on both footings. Under its post hoc amendment, on the mean of the two pairs, it reads AMPLITUDE CONSISTENT on both footings: KiDS-Legacy +1.46/+1.67 widths and DES Y3 +1.78/+1.76, with DES Y3 alt tagged inside the splice noise and KiDS-1000, reported, at +3.56/+3.69; with the ln-det term DES Y3 stays above 2 on both footings. It awaits an independent re-grade. LWB1, on DESI DR1 lenses, is NO VERDICT as frozen, since a known answer failed in the check, and AMPLITUDE CONSISTENT, FRAGILE under its post hoc amendment: σ_8 $0.786 \pm 0.029/0.787 \pm 0.029$, +0.90/+0.87 widths below form B's neutrino-floor value; it calls itself a consistency check, not independent evidence. Its blind re-grade, LWB1R, passes that check as frozen and so reads AMPLITUDE CONSISTENT, FRAGILE without an amendment (gap +0.91/+0.87); survey by survey KiDS-1000 has NO POWER, DES Y3 is in TENSION (+3.49/+3.47), HSC Y3 is consistent (−1.28/−1.32) and KiDS-1000 with DES Y3 is in TENSION (+3.12/+3.09). Its comparison reads AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL. It had seen LWB1's first-run rows before writing code, so it is not a fully blind re-grade, and the two lanes share their data and stored spectra. None of this closes item 12.
- a_0 **at** $z = 2.33$**.** AZ2 is NO VERDICT as frozen and FLAT HOLDS / EXPANSION-MODEL DEPENDENT under its second, post hoc amendment, conditional on Ω_m near 0.30; the lane says FLAT HOLDS is not a pass, since ΛCDM's constant dark energy gives the same, and its lead's re-run does not reproduce byte for byte. Its blind re-grade, AZ2R, reads FLAT HOLDS / EXPANSION-MODEL DEPENDENT on both footings: the headline cell holds (alt $\sqrt{X}$ 1.322), and the flat reading FAILS with BAO and supernovae at Ω_m 0.25 and 0.35, which is why the label is expansion-model dependent. Its comparison with AZ2's amended reading reads AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL; its continuous profile passes AZ2's frozen known answer as written, and it does not relabel AZ2's frozen NO VERDICT. It also says that a FLAT HOLDS is not a pass, since constant dark energy gives the same; that its new DESI pair is a PROVISIONAL reading of the source; that it shares AZ2's data, model and criteria; and that AZ2's amendment files, readable before its blind commit, already quoted the headline numbers. It closes no open item.
- **The Milky Way.** RAW1 grades the framework's line directly against DESI DR1's halo stars, by spherical Jeans modelling with no calibration and the adopted external field: FAILS HOT, FRAGILE on both footings, at census baryons and at the rotation-curve stellar mass. The failing bin is 30 to 40 kpc on every case (and 20 to 30 kpc too for census baryons on canonical). The lane says every smooth model fails hot in that bin, DESI's own NFW included, so it reads that bin as a feature of the stars, not of the framework alone; the FRAGILE tag flips only under a two-slope density whose outer slope is unconstrained; and its two amendments were each written after a failed run and before the run they govern. Two lanes then took that bin. HOT1 finds the hot 30 to 40 kpc stars CLUMPED, not localised and not a Sagittarius residue; the clumped stars are the bin's cold part, the heat is spread over the footprint, and the excess over the neighbouring bins is 2.25 sigma and may be a fluctuation.

LMC1 has NO LABEL as frozen (a precision check failed and a mutation did not bite); under an amendment written after its values were seen, which is not blind, it reads LMC NEGLIGIBLE: the LMC changes each bin's dispersion by less than the model's own noise and does not explain the hot bin. Neither relabels RAW1. DW1's frozen criteria award 9 of 12 checks; the extrapolated M_{200c} fails. NMB1: the line PASSES the shape-measuring DESI curves in all four cases, FRAGILE in one. NIM2 repeats that gate at three radii: with the field, the canonical line at census baryons FAILS LOW (2 to 3 sigma), FRAGILE, and every other case passes, three of them FRAGILE; the third radius adds amplitude, not shape power. PLT1: PASSES, FRAGILE with the Local Group field and PASSES without one; on a pre-named variant beyond 150 kpc the satellites FAIL LOW (median −2.72). SAT1 finds that the satellites' actual directions lower the field reading's apparent mass beyond 150 kpc, but not enough at census baryons (ANISOTROPY LOWERS IT, NOT ENOUGH). MWG1: hot gas does not supply the rotation curve (ROTATION CURVE NOT SUPPLIED), and with hot gas counted the outer rows pass, FRAGILE, on PROVISIONAL gas masses; HG2, with the published hot-gas laws, gives the same labels, with three models failing where MWG1 had one. MWJ1 is a post hoc consolidation with every input seen and reads CLOSES WITH CENSUS BARYONS, WEAKER FIELD, ROTATION CURVE INSIDE. MWI1, frozen and without amendment, asks whether published disc and bulge masses supply the stellar mass the rotation curve needs: SUPPLIED BY 1 OF 5 and NOT SUPPLIED at the record's census (canonical), SUPPLIED BY 2 OF 5 and WITHIN 2 SIGMA at the census (alt); the need is above every published central value, and the published masses are PROVISIONAL look-ups. RCF1 refits the rotation curve and finds that MWI1's need holds, with the classes changing on one of the two curves; STM1 reads the inventories from their full texts and finds MWI1's classes mostly hold, with the canonical count moving between 0 and 3 of 7 across the cold-gas range.

RG1, an independent re-grade whose numbers were saved before any lane script, output or result was opened, reads REPRODUCED for NMB1, PLT1, MWJ1, MWG1 and MWI1 (MWI1's geometry block not re-implemented). It shares the figure data and the baryon model with them, so it checks their arithmetic and grading, not their inputs. EFC1 then reads CONVENTION-DEPENDENT on both footings. The record's external-field value 0.01273, put inside the kernel as a Newtonian source, implies an actual field 9.4 (alt 10.2) times the acceleration it was derived from; EFC1 lists DW1's external-field rows and the field cases of PLT1, MWJ1, MWG1, SAT1 and NIM2 as using it. Under every other reading EFC1 tries, the outer data pass with census baryons, so PLT1's failing satellites, MWJ1's failing record point and MWG1's rescue by hot gas are statements about that one reading. EFC1 relabels none of these lanes and does not decide which reading is right. LIT1, a literature sweep that grades nothing, found no MOND or modified-gravity work on the Milky Way since DESI DR2 Results III (PROVISIONAL).

On 2026-11-12, after EFC1, the owner decided that the framework's lanes use the actual-field reading of the Milky Way's external field from here: e_N 0.000160 (canonical) / 0.000110 (alt) in place of 0.01273, with the field taking over near 800 kpc and the no-field line inside 300 kpc. The decision is forward only: earlier lanes stand as frozen and their field cases are read through it. Dated notes added to four lanes' files say what that means: in PLT1 the no-field rows are then the framework's, and the field rows and the FRAGILE finding belong to the earlier reading; in MWJ1 the adopted field lies inside the window at census baryons; in MWG1 Part 1 is unaffected and Part 2's rescue of the record's field is moot, with the no-field line at the gate on alt once hot gas is added and a very massive corona failing high; in DW1 the external-field rows, the plateau and the M_{200c} comparison belong to the earlier reading. The lead's argument for this reading is an argument, not a derivation, and no label changes. The same day the owner retired FALSIFIER_MATRIX row 32's first-registered mass kill, a measured $M(< 200\,\mathrm{kpc}) \geq 1.0 \times 10^{12}\,M_\odot$, together with the plateau prediction it guarded, because under the adopted reading the framework's own line gives $1.37/1.50 \times 10^{12}$ inside 200 kpc; the row stands with its re-scoped slope kill only (a slope of −2.25 or steeper over 100 to 200 kpc), ARMED, with no instrument yet. That is a kill test of ours retired after the data it would have judged had been seen, not a pass (Section 10, item xvii). The passes of NMB1, PLT1, MWG1 and MWJ1 are reproduced by an independent re-grade (RG1); those of NIM2, HG2 and SAT1's hot-gas variant await one.

- **Euclid strong lenses.** These grade the law acting on the lens galaxy's stars (law S). EUC1, on 3,192 lenses with every source at infinity: NO VERDICT on both footings, one-sided; the median lens sits at the ceiling. EUC4, on 124 wide lenses: WIDE LENSES BREAK THE CEILING on both footings, 75.8 % and 70.2 % above 1.78 times Salpeter; its blind re-grade EUC4R gives the same word and says its numbers are blind and its word is not. EUC2 (source redshifts from DESI fibres) and EUC5 (dynamical boost) are NOT EVALUABLE as frozen and under their amendments, and so is EUC3 (source redshifts from Euclid's grism tables), so EUC1's frozen gate stays open. EUC6 finds that wide lenses sit in richer environments (+3.38 galaxies, paired), which by its frozen reading places EUC4's excess with the group and cluster rows; its blind re-grade EUC6R gives

the same word (+3.59, the difference traced to how missing values were counted) and does not relabel it. EUC8 finds that the wide lenses need $3 \times 10^{11}\,M_\odot$ or more beyond the law on their stars inside 12.6 kpc, 45 to 61 % of the lensing mass; a compact central mass is TOO CONCENTRATED, every extended profile is allowed, and its word is SLOPE NOT CONSTRAINED. EUC5's blind re-grade EUC5R also reads NOT EVALUABLE under EUC5's amendment, and its comparison AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL; it records that commit subjects it had seen before its reading named EUC5's later readings, though no EUC5 number. EUC9 (Euclid's grism spectra) and EUC10 (published source redshifts) are NOT EVALUABLE: EUC10 finds 2 lenses of sample P with a published source redshift where the gate needs 10, so EUC1's gate stays open. The population gate EUC13 reads NO VERDICT on both footings, with its second statistic, f25, reported beside it as CEILING BROKEN; its blind re-grade EUC13R gives the same two words, its comparison AGREES WITH DIFFERENCES THAT DO NOT MOVE A LABEL, and it discloses that its blindness was incomplete. For lenses below 1.3 arcsec, two of three constructions put the class medians inside EUC5's sealed bands and the third at or just above the upper edge. EUC12, on one stellar-mass scale, reads NO VERDICT: Euclid's typical lens galaxies do not need measurably more stars than SLACS's. For the law on the stars, a hot-gas body with a core that would supply the wide lenses is excluded by its own pressure, so ONLY A CUSP IS LEFT (ESZ1), and that cusp is NOT EVALUABLE in X-rays as frozen and EXCLUDED only under a post hoc amendment (EXR1). Two lanes test a different kernel, the slow tail $1 - 1/\sqrt{1+2x}$, not form B's law: EUC7 reads INSIDE THE WINDOW on SLACS lensing and says this is not a pass of the kernel, since it adds an acceleration at 1 AU that the Solar System excludes and Euclid's wide lenses still break the ceiling; SLK1 reads RAR MARGINAL on SPARC.

- **The amount.** GAS1: MISSING BARYONS EXCLUDED [AMOUNT AND AMPLITUDE] on both footings (A3: SOUND WITH LIMITS). CLS2 and GO2 are in Section 5. U4L1, a secondary reading of an ALMA limit with no verdict weight, has NO LABEL as frozen and NO TENSION under its amendment; one disc excludes no model.

8.5 κ and the share

No lane derives κ or 5.41. FINAL16, FINAL-17, FINAL18, FINAL-19, FINAL6, OUT4 and OUT3's scale-free boundary each read KAPPA NOT DERIVED (FITTED) in their own words, and OUT3's cosmic-share fold reads AMOUNT NOT DERIVED (UNAVAILABLE), its construction missing by 18.5 band-widths. OUT-2's headline is that a clock-rate lock makes κ an output, $\sqrt{2/(3\pi)} = 0.460659$; audit A1 grades it WRONG: postulating the lock is algebraically the same as postulating that value of κ, so nothing is derived. A1 grades FINAL16 and FINAL-17 OVERSTATED (their verdict stands; their claims of criteria committed alone are false, and FINAL-17's capacity is one line of the law's algebra), FINAL18, FINAL-19 and OUT3's fold SOUND WITH LIMITS, and OUT4 SOUND. C3 adds that OUT4's dimension census shows only that dimensional analysis leaves κ free, which is not a no-go, and that FINAL-17's reading of κ as a flat direction does not stand, since at fixed baryon mass and radius a_0 is observable. The Lean lanes certify arithmetic in which 5.41 is an input (FINAL-11, FINAL-14) or do not compile (FINAL-10); FINAL-12's committed result reports a compile at exit 0, and A1 records that the copy it built is untracked, differs in its bytes and fails. Of the formation lanes, FINAL2 reads NO POWER, FINAL15 FAILS as frozen and holds only after a correction of its own specification, FINAL7F reads NO STRUCTURE as frozen, and FINAL6 reads KAPPA NOT DERIVED (FITTED) + FORMATION REFUTED (per-galaxy); an independent audit of 2026-11-12 (C3) finds that its step selecting ρ_Λ does not hold, since its own test uses a density frozen at a fixed epoch, which is also constant, so FINAL6 does not pick the canonical footing, and its refutation covers the two per-galaxy constructions it tested; A2 grades FINAL15 and FINAL7F OVERSTATED, and FINAL15's holding reading awaits an independent re-grade. FINAL5 proposes a stacked thermal SZ measurement and grades nothing; A2 finds it not decisive for the joint, and C4 finds that its detectability forecast does not stand: it took the column as a flat sheet, used a beam that is not a two-dimensional convolution and overstated the X-ray flux by 10^4.

8.6 Every lane folded into version 1.2

Labels canonical / alt as each file states them; "both" means the same label on both footings. "awaits" means awaits an independent re-grade.

Lane	Question	Label as frozen	Amendments (post hoc)	Re-run, blind re-grade, audit
The forest				
FRZ1	forest row with the redistribution at z 2.2–4.4	FOREST ROW FAILS / FOREST ROW FAILS	none	lead's re-run reproduces; blind build FRZ1R
FRZ1R	blind build of FRZ1's row	FOREST ROW FAILS (8142.08) / FAILS (7685.51)	none	comparison: agrees
FRZ2	forest with $a_0(z)$	NO FRAMEWORK $a_0(z)$ PASSES	none	
FZS1	law's layout only inside the settled radius	FOREST ROW FAILS, FRAGILE (both)	the rule is post hoc to FRZ1	
SRH1	one settled-radius history, forest against lensing	WINDOW EMPTY / WINDOW EMPTY	A1: tilt cases, reported, never graded	re-run reproduces; lensing gate re-graded blind (SRH1R)
SRH1R	blind re-grade of SRH1's lensing gate	NO VERDICT / NO VERDICT	one, before data	does not relabel SRH1
FRZ3	the forest row at flux level	NO VERDICT / NO VERDICT	A1, A2, before any graded number	lead's re-run of the grader reproduces
FRZ3R	blind flux-level build	NO VERDICT / NO VERDICT; reported reading T: FOREST ROW FAILS AT FLUX LEVEL	A1 (reading T), before any flux number	blindness breach disclosed; lead's re-run of the grader reproduces
HLF1	CO1's rows on the own halo list; forest row	NO VERDICT / NO VERDICT	A1: Part A ROWS HOLD WITH THE OWN LIST	awaits
FINAL4	growth history of the formation reading	DERIVED HISTORY EXCLUDED (both)	tilt reading post hoc	
The joint and the medium				
SET1	irreversible settling to the law's pressure	STRUCTURE FAILS AT A1 (both)	none	
EDG1	charge held in an edge layer	FAILS AT E3 / FAILS AT E3	none	
AKS1	acceleration-keyed sink; local dissipation	control as recorded; Part T CLOSED, ONE BRANCH ON AN ESTIMATE (both)	Part T is post hoc; A1	
VSH1	AKS1's estimate T7, computed	NO VERDICT / NO VERDICT	A1: SETTLES BY SHOCK (both)	not re-run; awaits; A3 SOUND WITH LIMITS
VSJ1	is VSH1's settled state the joint	NO VERDICT / NO VERDICT	A1: STANDS BUT MISCOUNTED (both)	not re-run
VCC1	can the clock's charge be sourced	EXCLUDED / EXCLUDED	none	A3 SOUND
CLOSE2	PAR1's two open branches	NO VERDICT / NO VERDICT	CANDIDATE STRUCTURE FAILS (both)	A2 SOUND WITH LIMITS
CLOSE-1	engineered kinetic function	FOLDS / FOLDS	none	A2 SOUND WITH LIMITS; C3: linear gate now fails, label stands
PIK1	dust term gated on the halo density	NO VALUE / NO VALUE	none	no row computed; C2: NO VALUE not established; rows computed by PIK2
PIK2	the gated term's rows, computed	CMB-ONLY / CMB-ONLY	none	blind; lead's re-run reproduces
A0Z1	the χ sector and $a_0(\chi)$	EOM CERTIFIED; $a_0(\chi)$ FAILS (both)	none	
FRM1	the formation reading as a theory	frozen line FAILS / FAILS; reported HOLDS EXCEPT (both)	one, before numbers	awaits; C2: bands retrospective
FINAL-1	the theory assembled	THEORY WRITTEN (both)	none	A2 OVERSTATED
FINAL-7	what the medium is	NO MEDIUM, CLOSED AT (ii) (both)	none	A2 SOUND WITH LIMITS
ELA1	an elastic solid as the medium	ELASTIC SOLID FAILS AT S0 (+S2, B1) (both)	A1	A2 SOUND
STR1	can the anisotropic stress pay the lensing	STRESS CANNOT PAY (both)	none	A3 SOUND WITH LIMITS
OUT5	a pinned bootstrap carrier	BOOTSTRAP KILLED (FRW/HALO + PAYMENT)	none	A2 SOUND WITH LIMITS
NBC1	contact with neutral gas	KILLED (K2 PROVISIONAL) (both)	none	
TOE0	gate battery for a theory of everything	NO TOE, in either sense	A1, before any number	
QG1	quantum health of action (A)	T2 FAIL: Minkowski FAILS (both)	A1, before the graded run	

Lane	Question	Label as frozen	Amendments (post hoc)	Re-run, blind re-grade, audit
HZ1	a UV completion of action (A)	T3 FAIL	A1 (one MUTATE twin)	
NPJ1	a non-perturbative origin of J	SHAPE ONLY	none	
LSF1	a Lifshitz fermion superfluid	KILLED AT G0 (both)	none	
External field				
EFN1	nonlinear outcome of SCS3's instability	NO VERDICT / NO VERDICT	A1–A4: NO VERDICT	
Data rows				
BX1	pair B finished (addendum)	NO VERDICT / NO VERDICT (unchanged)	A1: AMPLITUDE CONSISTENT (both)	awaits
LWB1	amplitude on DESI DR1 lenses	NO VERDICT / NO VERDICT	A1: AMPLITUDE CONSISTENT, FRAGILE	blind re-grade LWB1R
LWB1R	blind re-grade of LWB1	AMPLITUDE CONSISTENT, FRAGILE (both)	none	not fully blind (its file)
AZ2	flat a_0 at z 2.33, DESI DR2 Lyα	NO VERDICT / NO VERDICT	A2: FLAT HOLDS / EXPANSION-MODEL DEPENDENT	re-run not byte-identical; blind re-grade AZ2R
AZ2R	blind re-grade of AZ2	FLAT HOLDS / EXPANSION-MODEL DEPENDENT (both)	none	blindness caveat (its file)
GO2	group census on DESI groups	NOT EVALUABLE [r_ta NOT REACHED] (both)	none	lead's re-run reproduces
CLS2	what the cluster mass is	NO CANDIDATE / NO CANDIDATE	none	
CRACK-1	three readings of the ultra-faint and X-COP pair	FAILS THE PAIR / FAILS THE PAIR	A1; same labels	A3 SOUND WITH LIMITS
GAS1	is the amount missing baryons	MISSING BARYONS EXCLUDED (both)	none	A3 SOUND WITH LIMITS
RAW1	the halo stars directly	FAILS HOT, FRAGILE (both)	A1, A2, each before its run	
HOT1	where the hot bin's stars are	CLUMPED	none	
LMC1	the LMC's pull on the halo stars	NO LABEL; A3 reading LMC NEGLIGIBLE	A1–A3; A3 not blind	
DW1	DESI DR2 Milky Way mass profile	9 of 12 checks; M_{200c} fails	none	dated note
NMB1	line against DESI's NIMBLE curves	PASSES (E1 canonical FRAGILE)	none	RG1: reproduced
NIM2	NMB1's gate at three radii	E1 canonical FAILS LOW, FRAGILE; others PASS	none	awaits; EFC1
PLT1	outer masses at 100 kpc and beyond	PASSES, FRAGILE (field); PASSES (no field)	a post hoc reading	RG1: reproduced; EFC1; dated note
SAT1	the satellites' actual directions	ANISOTROPY LOWERS IT, NOT ENOUGH (both)	A1, before any label	EFC1
MWG1	hot halo gas	ROTATION CURVE NOT SUPPLIED; outer rows PASS, FRAGILE	none	RG1: reproduced; EFC1; dated note
HG2	published hot-gas laws	same labels as MWG1, FRAGILE	none	awaits
MWI1	published stellar masses against the inner need	SUPPLIED BY 1 OF 5 / SUPPLIED BY 2 OF 5; census NOT SUPPLIED / WITHIN 2 SIGMA	none	RG1: reproduced
RCF1	the need under a rotation-curve refit	MWI1's need HOLDS	A1, a diagnostic only	
STM1	the inventories from their full texts	MWI1's classes mostly hold	A1, before any z	
MWJ1	one Milky Way window	CLOSES WITH CENSUS BARYONS, ...	post hoc consolidation	RG1: reproduced; EFC1; dated note
RG1	independent re-grade of five Milky Way lanes	REPRODUCED (all five)	none	shares inputs
EFC1	the external-field convention	CONVENTION-DEPENDENT (both)	a post hoc consolidation	
LIT1	Milky Way prior art	a literature sweep, no label	none	
EUC1	Euclid lenses at the ceiling	NO VERDICT / NO VERDICT	none	

Lane	Question	Label as frozen	Amendments (post hoc)	Re-run, blind re-grade, audit
EUC2	source redshifts in DESI fibres	NOT EVALUABLE	A1: NOT EVALUABLE	
EUC4	wide Euclid lenses	WIDE LENSES BREAK THE CEILING (both)	none	blind re-grade EUC4R: same
EUC4R	blind re-grade of EUC4	WIDE LENSES BREAK THE CEILING (both)	none	word not blind (its file)
EUC3	source redshifts from Euclid's grism tables	NOT EVALUABLE / NOT EVALUABLE	post hoc diagnostics	
EUC6	wide-lens environment	WIDE LENSES SIT IN RICHER ENVIRONMENTS	none	blind re-grade EUC6R: same
EUC6R	blind re-grade of EUC6	WIDE LENSES SIT IN RICHER ENVIRONMENTS	none	does not relabel EUC6
EUC8	the wide lenses' residual profile	SLOPE NOT CONSTRAINED / SLOPE NOT CONSTRAINED	C1, before any script	
EUC7	slow-tail kernel on the lens gates	INSIDE THE WINDOW / INSIDE THE WINDOW	none	not a pass (its file); awaits
SLK1	slow-tail kernel on the RAR	RAR MARGINAL / RAR MARGINAL	none	
EUC5	dynamical boost of lens galaxies	NOT EVALUABLE / NOT EVALUABLE	A1: NOT EVALUABLE	blind re-grade EUC5R
EUC9	source redshifts in Euclid's grism spectra	NOT EVALUABLE / NOT EVALUABLE	A0 before data; A1 post hoc: NOT EVALUABLE	
EUC10	published source redshifts	EUC1-F NOT EVALUABLE (both)	none	
EUC12	a common stellar-mass scale	NO VERDICT / NO VERDICT	none	
EUC13	population source gate	EUC13-P NO VERDICT (both); f25 reported: CEILING BROKEN	A1, post hoc	blind re-grade EUC13R: same
EUC13R	blind re-grade of EUC13	NO VERDICT (both); f25 CEILING BROKEN	none	blindness incomplete (its file)
ESZ1	wide-lens gas pressure	ONLY A CUSP IS LEFT (both); Part C NO EXCESS DETECTED	none	
EXR1	wide-lens X-ray cusp	NOT EVALUABLE / NOT EVALUABLE	A1, post hoc: EXCLUDED	
EUC5R	blind re-grade of EUC5	A1 as frozen: NOT EVALUABLE (both)	F1S, post hoc	does not relabel EUC5
U4L1	ALMA limit on one disc	NO LABEL	A1: NO TENSION	no verdict weight
κ and the share				
FINAL16	is κ an output	KAPPA NOT DERIVED (FITTED), FORM OUTPUT	A1 (J7), before the run	A1 OVERSTATED
FINAL-17	κ from the capacity	KAPPA NOT DERIVED (FITTED), CAPACITY DERIVED	A1, after the outputs (audit A1)	A1 OVERSTATED
FINAL18	κ from matching at the fold	KAPPA NOT DERIVED (FITTED), FOLD MATCHING FALSIFIED	none	A1 SOUND WITH LIMITS
FINAL-19	κ from the deep-field column	not derived; the column is 6.0/7.3 times too large	none	A1 SOUND WITH LIMITS
FINAL6	κ from the formation reading	KAPPA NOT DERIVED (FITTED) + FORMATION REFUTED	none	
OUT-2	a clock-rate lock	$\kappa = 0.460659$ claimed as an output	none	A1 WRONG
OUT3 (fold)	the amount from the fold	AMOUNT NOT DERIVED (UNAVAILABLE)	none	A1 SOUND WITH LIMITS
OUT3 (boundary)	$\kappa = \frac{1}{2}$ and the vacuum value	NOT DERIVED (FITTED)	A1, before the Lean run	A2 SOUND WITH LIMITS
OUT4	the κ window	KAPPA NOT DERIVED (FITTED); WINDOW 9.2x TOO WIDE	none	A1 SOUND
FINAL2	which epoch fixes the amount	(e) BEST-RANKED / NO POWER (both)	none	
FINAL15	the formation ledger	FAILS / FAILS	specification defect D1 corrected: HOLDS / HOLDS IN PATTERN, NOT IN SIZE	A2 OVERSTATED; awaits

Lane	Question	Label as frozen	Amendments (post hoc)	Re-run, blind re-grade, audit
FINAL7F	residual as history or present state	NO STRUCTURE / NO STRUCTURE	A1: NOT A FUNCTION OF ANY TESTED VARIABLE	A2 OVERSTATED
FINAL5	a proposed observation	a proposal, no label	none	A2 OVERSTATED
FINAL-10/11/12/14	Lean certificates	FAIL; CERTIFIED; exit 0 (30 theorems); six items compile	none	A1: see text

9 Version 1.3: a candidate sixth rule, its scorecard and the rows since version 1.2

The lanes in this section landed after version 1.2 was deposited, on 2026-11-12 and 2026-11-13. Each froze its criteria in a commit before any graded number unless its row says otherwise; frozen readings come first and post hoc readings are labelled. The completed form and its scorecard are taken from the assembly CJ1 (`closure_2026_11_13/CJ1_candidate_completion/`, `COMPLETED_FORM.md` and `SCORECARD.md`), which computes nothing new, freezes nothing and relabels nothing, and whose own audit checks every quoted word against its committed file (116 of 116). The words for the completed form in its scorecard are the lead's placements of each lane's committed words, made the same way in both directions. A HOLDS, a PASSES or a hit that no independent blind re-grade has confirmed is called a hit here.

9.1 The candidate sixth rule: the crossing-speed joint (NOT ADOPTED)

Rules 1 to 5 are those of Section 7.1. CJ1 fills rule 6 with the crossing-speed joint as lanes JNT2 and JNT2R built it. It is a completed statement, not a confirmed theory, and it is NOT ADOPTED: adopting it is the owner's decision.

6. **The joint (CANDIDATE: the crossing-speed joint).** The law's layout holds inside $R_{\rm set} = f \times R_{\rm fold}$ in systems slower than

$$v^*(z) = v^*(0)\,\frac{H_0}{H(z)}, \qquad v^*(0) = \frac{a_0}{10\,(1+X)\,H_0} = 668.5/807.9\ \mathrm{km\,s^{-1}}\ (669/808\ \text{rounded}),$$

with $X = 5.41$ the share. Faster systems keep their share as cold dust in the reference layout (the record's cold reference profile) at every radius. Beyond $R_{\rm set}$ in slow systems the cold dust also keeps the reference layout, joined at $R_{\rm set}$ by FZS1's junction. $R_{\rm fold}$ is FLD1's fold radius at the system's own baryon mass and redshift, and f is a constant. A system's speed is the virial speed $v_L = (10\,G\,(1+X)\,M_b\,H(z))^{1/3}$; the measured speed is a bracket. Equivalently, a system is above the switch if and only if $M_b > M_b^*(z) = v^*(z)^4/(Ga_0)$: $\log M_b^*$ is 13.21/13.45 today, 12.19/12.44 at $z = 1$ and 11.28/11.53 at $z = 2$. $H(z)$ is flat with Ω_Λ 0.6847 and H_0 67.4, and a_0 is held constant in z.

Where v^* comes from. Route D's check C5 (`fable52_push/SHAPE_search/route_D_dark_energy_column.md`): cold clustering gives $v^3 = 10GMH$, the virial relation at 200 times the critical density, with $M = (1+X)M_b$, while the law's flat speed gives $v^4 = GM_ba_0$; with the cosmic share the two meet at one speed only. Below that speed the share a system brings exceeds what the law's halo uses; above it the law's halo would need more than the whole share, which rule 5 forbids. So no new constant enters v^*: it follows from rules 2, 4 and 5 and the virial definition, and it still carries both fitted constants, κ through a_0, and X.

Number	Value (canonical/alt)	Status	Where it comes from
κ	$\frac{1}{2}$	FITTED	rule 4; KAP1 and OUT4: KAPPA NOT DERIVED
X, the share	5.41 (5.411/5.412 in CO2)	FITTED	rule 2; R541A to R541E: four EXCLUDED, one TRADES
$v^*(0)$	$668.5/807.9\,\mathrm{km\,s^{-1}}$	FOLLOWS FROM THE RULES	the lemma above; recomputed in SI by JNT2 and JNT2R
$v^*(z)$	$1/E(z)$: 373.3/451.1 at z 1, 220.4/266.4 at z 2, 177.4/214.4 at z 2.5	FOLLOWS FROM THE RULES	the lemma with $H(z)$ in place of H_0; a_0 constant

Number	Value (canonical/alt)	Status	Where it comes from
f	ungraded window [1.25, 2] in both builds on both footings; JNT2 alone reads [1.25, 3] canonical	BOUNDED BY DATA, NOT FIXED	see below
$R_{\rm fold}$	e.g. 169 kpc today at $\log M_b$ 10.7 (canonical; alt within 0.5 %)	TAKEN FROM A LANE'S CONSTRUCTION	FLD1

f, edge by edge. Both builds report the window as ungraded, with the coverage line, which fails, set aside. The lower edge, 1.25 in both builds and on both footings, is set by KiDS early-type lensing at 1.6 R_h, which fails at $f = 0.75$ and 1 ($+2.003/+2.037\sigma$ at $f = 1$) and passes from 1.25; the blind re-grade of that gate, SRH1R, found that it passes at $f = 0.05$ and with the reference everywhere, so the gate cannot see f (Section 5, item 10). The upper edge is the forest at matter level at $f = 3$: JNT2 reads 1.99 (holds) canonical and 6.43 (fails) alt, and JNT2R reads 17.6/29.2 (both fail), so the upper edge is 2 in JNT2R on both footings and in JNT2 on alt; JNT2's canonical 3 is not agreed, the cause of the difference was not traced, and f between 2 and 3 was not run. At flux level FRZ6 finds that the forest does not cap f at 2 (Section 9.3); the blind FRZ6R reads NO VERDICT as frozen on every split case, with $f = 3$ under the gate except at the halved smoothing scale, so this is a hit that the blind build neither confirmed as a label nor contradicted.

$R_{\rm fold}$, and what FLD1 assumes. FLD1 (label DOES NOT/DOES NOT) follows independent radial shells of a clock that starts as cold dust in the Hubble flow at $z = 10$ around a softened point mass of baryons, in the law's static field. Its limits are its own: spherical symmetry; a shell equation exact only up to the first crossing, behind which the action gives no answer (COL1); no discs, gas or feedback. $R_{\rm fold}$ is about 0.52 of the turnaround radius today and grows roughly as $M_b^{1/3}$.

What is not derived. The two constants: $\kappa = \frac{1}{2}$ and $X = 5.41$ are fitted, no lane derives either (KAP1, OUT4, R541A to R541E), and v^* inherits both. Why the settled region is f times the fold radius: the settled-radius family was chosen after the forest row failed (FRZ1), so it is post hoc, and f is read from data, not computed. How a falling clock becomes a standing one: rule 6 says where each layout holds; it is a rule, not an equation, and no settling route tried reaches the static state (COL1, SET1, AKS1, VSH1, VSJ1, EDG1, CLOSE-1). Why the boundary is sharp: nothing derives a sharp switch or its width. The medium is not identified and the kinetic function is not fixed. Every open item of Section 5 that rule 6 does not touch stays open. The crossing-speed boundary is itself post hoc to the earlier split at FRZ1's constant speed (SRJ2) and to JNT1.

Prior work. A provisional web search by the lead session, read through a summarising tool (PROVISIONAL; it is not a lane and no committed file holds it), finds the ingredients published. The galaxy relation of slope 4 set against the virial relation of slope 3 is the origin of the detected baryon fraction falling for smaller systems, with clusters close to the cosmic fraction, and inside R_{500} that fraction is proportional to the circular speed, as the two relations imply (Famaey and McGaugh 2012, arXiv:1112.3960, Sections 4.2, 4.3.1 and 5.2). A broken relation with a cluster branch above about 350 km s^{-1} has the detected baryon fraction below unity in every system, with no formula for the speed at which it would reach unity (McGaugh, Schombert, de Blok and Zagursky 2010, arXiv:0911.2700). Clusters are the scale where nearly all the expected baryons are detected and where systems stop resembling MOND and start resembling a cold-halo cosmology, with no intersection speed given (McGaugh 2015, arXiv:1404.7525, Section 4). Groups up to about 200 km s^{-1} follow the deep-MOND relation, with no upper limit proposed (Milgrom 2018, arXiv:1810.03089 and arXiv:1811.12233). Superfluid dark matter switches between galaxies and clusters by temperature and a core radius, not by a_0, the baryon share and the Hubble rate (Berezhiani and Khoury 2015, arXiv:1507.01019; Hodson et al. 2017, arXiv:1611.05876). We did not find the crossing written as one speed $a_0/(10(1 + X)H_0)$ fixed by the cosmic share, its $1/H(z)$ scaling, its reading as a rule that switches layouts, or its application to massive discs at $z \approx 2$; in that narrow sense it appears new. It is a corollary of two published relations and the cosmic share; what the record adds is its use as a rule with a redshift scaling and the predictions sealed from it (Section 9.5). The search was not complete: the body of Marongwe and Kauffman 2025 (arXiv:2511.20188; only its abstract was read, which describes one relation whose normalisation evolves with cosmic time), the cluster section of Famaey and McGaugh, McGaugh's papers of 2011 and 2012 and the papers citing Milgrom 2017 (arXiv:1703.06110) were not checked.

9.2 The completed form against every tested row

CJ1's scorecard grades nothing new: each of its 45 rows quotes its lanes' committed words, frozen word first, and then places the row against the completed form with one of a fixed list of words. The counts are the same on both footings.

Word for the completed form	canonical	alt
EQUALS THE LAW (rule 6 gives the law's own layout there)	14	14
NOT GRADED FOR THIS FORM	8	8
NO POWER	6	6
FAILS	5	5
PASSES TRIVIALLY (the form equals the reference there)	4	4
DEPENDS ON AN UNTESTED ASSUMPTION	4	4
PASSES	3	3
FAILS ON COVERAGE ONLY	1	1
Total	45	45

The rows that carry the story:

- **The one frozen fail.** JNT2 and its blind twin JNT2R both read JOINT FAILS ON G2 (coverage): 10/2 discs of RC100 at z 2.10 to 2.52 sit above $v^*(z)$, so the law covers 90/98 of 100 at every f where the coverage line asks for it. The fail is on coverage, not data: one point per disc cannot tell the reference, half of it or the law apart (SRJ3, SRJ4: NO VERDICT, no power). Doubling v^* would restore coverage and break the forest, so rescuing the coverage line needs a fitted constant between 1 and 2 v^*.
- **Trivial passes and the law's passes.** The passes are TRIVIAL (the forest at matter level up to f 2, X-COP's one-sided amount and its radial shape under RB1's post hoc amendment, the CMB rows) or EQUAL TO THE LAW (SPARC, where 0 of 175 galaxies are above v^*, KiDS, the Milky Way, the groups). No row that passes is support for the boundary itself.
- **The three PASSES, none confirmed by a blind re-grade.** KiDS early-type lensing at $f \geq 1.25$, a gate SRH1R found cannot see f; the forest at flux level for the split, FRZ6's hit, which its blind re-grade FRZ6R neither confirmed as a label nor contradicted (Section 9.3); and Coma's ultra-diffuse galaxies with the Milky Way's classical dwarfs, inside the interval only with the systematics budget, which rule 6 does not touch.
- **The resolved discs.** DSC1's hit above v^* is no longer listed: the blind DSC1R reads NO POWER there, and below v^* the two builds read opposite words (Section 9.4). The nine massive SPARC spirals are all below v^*, so the candidate gives them the law, which fits, fragile, on canonical in the blind JNT1R; the earlier split at a constant speed gave them the reference, which fails (JNT1R: REFERENCE FAILS AT THE RECORD SHARE).
- **The fails.** The Milky Way's halo stars at 40 to 55 kpc once the star-count slope is the measured RR Lyrae one, fragile and shared by every smooth model (RRL1); and, untouched by rule 6, form B's fails on X-COP's two-sided amount, the outer cluster shear, the ultra-faints and M31's bright dwarfs, and DF2, DF4, NGC 147 and IC 1613.

No hit of the completed form has had an independent blind re-grade that confirmed it; the blind re-grade of FRZ6 reads NO VERDICT as frozen and so does not confirm its hold, while its numbers sit under the gate and do not contradict it. CJ1 keeps row 22's word by the rule it already used in row 11 and marks it unconfirmed.

9.3 The forest at flux level

The law's full layout FAILS. FRZ4 realised the rare haloes bin by bin and graded the row at flux level with no amendment: FOREST ROW FAILS AT FLUX LEVEL on both footings, $\Delta\chi^2$ 269/288 against the gate of 4, and no bracket brings it under 4 (the lowest 59/38). This is the first graded flux-level label for the forest row. Its matter-level known answer passes in every case. The settled layout at f 1, 2 and 3 is NO VERDICT (known answer e fails): two seeds of the base box differ by as much as the effect itself.

The gate fits no gas parameter. FRZ5 found that the record's gate is an Asimov $\Delta\chi^2$, not a fit: the model inside it has no forest nuisance parameter at all, so it treats the forest's gas physics as exactly known. As frozen, FRZ5 reads NO VERDICT (a twin does not bite) on both footings. Under its amendment A1, post hoc and committed alone before its numbers, it reads FOREST ROW FAILS AFTER REFIT on both footings. Letting

a per-bin amplitude, a tilt and a smoothing float removes about 91 % of the full layout's flux-level $\Delta\chi^2$: 269/288 falls to 23.0/23.9, about twelve-fold, still about six times the gate, with no bracket bringing it under 4 (lowest 21.5). The matter-level number the record first used (9916/9009) refits to 118/107 only with the smoothing pinned at its floor.

The candidate's split HOLDS at two settled radii, as frozen; a hit. FRZ6 ran the crossing-speed split at flux level with 12 seeds at $z = 2.2$. Its frozen criteria were committed alone before any script and deposited before any result, DOI 10.5281/zenodo.24208901 (created 2026-11-13 22:41 UTC); the result is version 2 of that record, DOI 10.5281/zenodo.24210556 (concept DOI 10.5281/zenodo.24208900), with its criteria, amendment, scripts, part files and graded outputs. As frozen: at $f = 2$ FOREST ROW HOLDS AT FLUX LEVEL canonical and FOREST ROW HOLDS AT FLUX LEVEL, FRAGILE (lambda_P x 0.5) alt; at $f = 3$ the same FRAGILE word on both footings; $\Delta\chi^2$ 0.196/0.257 at $f = 2$ and 1.144/1.434 at $f = 3$, and with twice the seed error added 0.50/0.65 and 2.04/2.50, against the gate of 4. Every frozen caveat goes with it:

- **Fragile on three of four.** Halving the pressure-smoothing scale pushes $f = 3$ on both footings and $f = 2$ alt over the gate. Under the published measured smoothing range (a PROVISIONAL reading of Rorai et al. 2017, reported and never the label) nothing crosses.
- **At $f = 2$ the number is a noise floor.** The flux change there is below what 12 seeds resolve; the HOLDS says the change is under the gate, not that a signal was measured. $f = 1.25$ is NO VERDICT (known answer e fails): even 12 seeds do not resolve a change that small.
- **The control rests on a post hoc amendment.** As frozen, the unsplit layouts and the full-layout control are NO VERDICT because a known answer demanded that FRZ4's 3D matter power be reproduced to 10^{-12}, which single precision cannot meet (1.7e-7); the graded flux statistic is reproduced to 6.7e-16. Only under AMENDMENT_A1, post hoc and committed alone, does the control reproduce FRZ4's FOREST ROW FAILS AT FLUX LEVEL (256.1/282.3; 22.0/21.3 after FRZ5's refit), and the unsplit layout at $f = 3$ hold, FRAGILE.
- **Not confirmed.** Every HOLDS is a hit, not a pass of the theory: the settled-radius family and the crossing-speed boundary are post hoc, and κ and 5.41 are fitted. A hold here means the layout is not excluded, not that it is preferred. The lead re-ran FRZ6's grader and refit from a copy and all twelve outputs reproduce byte for byte; that is a reproduction, not an independent re-grade.

The blind re-grade (FRZ6R): NO VERDICT as frozen; it neither confirms nor contradicts the hold. FRZ6R built the same split blind and independently: its own split tables, realisation, seeds, estimator and grader, with FRZ3R's FGPA functions reused unchanged and 4 seeds per bin in all 12 bins. Its criteria were committed alone before any script; its amendment A1, committed alone before any graded number, adds a relative bite rule for one twin as a reported reading. As frozen it reads NO VERDICT for every case on both footings, $f = 3$ with FRAGILE (lambda_P x 0.5), and its control (the law's layout everywhere) FAILS (D 323.0, SE 4.5 canonical; D 395.8, SE 8.5 alt), against FRZ3R's reading T of 363.26/424.50, that is $-11\,\%/-7\,\%$. The only failing item is twin TW1, which removes the boundary: it does not bite under the frozen rule nor under A1's relative rule (a change of 0.142 against twice its seed error, 0.150, canonical; 0.068 against 0.091 alt), so no case can be HOLDS. Every known answer passes, twin TW2 bites, and D + 2 SE is under 4 in every split case.

Case	canonical D (SE)	alt D (SE)
$f = 1.25$	0.00056 (0.00024)	0.00130 (0.00054)
$f = 2$	0.0244 (0.0032)	0.0293 (0.0035)
$f = 3$	0.572 (0.044)	0.691 (0.047)
control	323.0 (4.5)	395.8 (8.5)

Only one bracket crosses the gate: $f = 3$ at lambda_P x 0.5 (40 comoving kpc), 4.29/4.29, hence FRAGILE; the lane itself cautions that at that smoothing its grid does not resolve the scale, so that bracket is also the least reliable one. Under the measured-range reading (lambda_P 91.4 to 138.6 comoving kpc, PROVISIONAL, reported) $f = 3$ reads D 0.057 to 0.58 canonical and 0.072 to 0.64 alt, still NO VERDICT (twin TW1), and the control 112 to 247 and 136 to 300, failing. The lead re-ran its grader from a copy and the four outputs equal byte for byte; the boxes were not re-run. FRZ6R's criteria, amendment, result and files are version 3 of FRZ6's Zenodo record, DOI 10.5281/zenodo.24211910 (created 2026-11-14 03:37 UTC; concept DOI 10.5281/zenodo.24208900); version 1, DOI 10.5281/zenodo.24208901 (2026-11-13 22:41 UTC), remains the pre-registration timestamp of FRZ6's criteria. So the blind build does not confirm FRZ6's HOLDS as a label, and it does not contradict it: its numbers are lower than FRZ6's and under the gate, and the full layout fails in both builds. FRZ6's hold stays a hit, not a confirmed hold. In FRZ6R's own words, a firmer word needs a twin that bites, for example more seeds

for the twin or a twin defined at full 12-bin scope, frozen in advance, and even then a HOLDS here would be a hit, not a pass of the theory.

Left out, as in FRZ4: hydrodynamics, shocks and hot halo gas, self-shielding, fluctuations of the ultraviolet background and temperature, peculiar velocities, and a box larger than 20 Mpc/h; only $z = 2.2$ has 12 seeds. Beside these, the unsplit settled family's own flux window, SRF1 and its blind twin SRF1R, is NO VERDICT/NO VERDICT as frozen in both builds; SRF1R's ungraded window is [1.004, 3.16]/[1.050, 3.21].

9.4 The data teams' assumptions

The Milky Way's hot bin. RAW1's hot 30 to 40 kpc bin (Section 8.4) was taken apart. MWT1 asks what the fail rests on: as frozen, FAIL DEPENDS ON AN ASSUMPTION THE DATA CANNOT TEST, the tracer density beyond the sample's reach, at census baryons on canonical (on alt also the distance scale); on the rotation-curve stellar mass on alt the fail is removed by the logg cleaning or by equal hemisphere weighting, each alone, FRAGILE, a hit at the noise level that awaits an independent re-grade. Under the combination of everything the data detect the bin stays HOT in all four cases. MWT2 asks whether the bin is a structure and reads NO POWER in every case: a planted group of the right size is found but not called distinct, so a "not distinct" carries no weight; as information, the 30 to 40 kpc bin is the only one with two components, the second cold and holding 12 per cent of the stars. RRL1 measured the halo's density slope from Pan-STARRS1 RR Lyrae: a break is detected, slope 5.293 inside 39.0 kpc and 3.934 beyond. Redone with that slope, RAW1's test reads MEASURED SLOPE MAKES A PASSING BIN FAIL [40-55 COLD] in all four cases, FRAGILE in three (robust only on the rotation-curve stellar mass, alt): the hot bin passes (census 1.205 to 1.052 canonical, 1.155 to 1.008 alt) and the 40 to 55 kpc bin turns cold; DESI's NFW moves exactly like the framework's line. The swap moves every smooth model the same way and singles out no model.

The resolved discs at $z \approx 1$ to 2.4. DSC1 graded the candidate on digitised outer rotation curves: above v^*, JOINT'S STATEMENT HOLDS ON THE RESOLVED DISCS, FRAGILE (canonical) and NO POWER on alt; below, LAW FAILS BELOW, FRAGILE on both footings. The blind DSC1R reads NO POWER, FRAGILE above and LAW HOLDS BELOW, FRAGILE below, on both footings, and every layout, the law included, is rejected in absolute terms. So the blind re-grade confirmed neither the hit nor the fail. Its reconciliation, written after its blind work was committed, traces the difference above v^* to how many digitised points count as independent: DSC1 counts every point, DSC1R one per 0.6 arcsec beam; at the record's halo DSC1 reads A 0.670 [0.250, 1.175] and DSC1R 0.590 [-0.13, 1.58] (canonical). Below v^* no ingredient changes which layout wins: a free reference halo at A about 1.25 to 1.57 fits better than the law in both builds, and the two words differ only by the decision line. ALPAKA's ring curves read NO POWER in ALP1 and its blind twin ALP1R.

The Euclid lens gate. EUC16, with a better arc finder, certified 11 source redshifts where the gate needs 10 and graded EUC1's source-redshift gate for the first time: NO VERDICT/NO VERDICT. The median boost of the law on the stars is +0.405 [+0.137, +0.589] canonical and +0.385 [+0.121, +0.574] alt, against a hold line of 0.15 and a fail line of 0.25: the centre sits on the failing side, and the interval is too wide to decide. The certified lenses are not typical: they read about 0.13 heavier than the whole sample before any source redshift. The candidate changes nothing here: 26/0 of 6,173 Euclid lenses are above $v^*(z)$ (LNJ1), so its lens prediction is the law's, and the wide lenses' ceiling break is not its switch (LNJ1: NO POWER, FRAGILE/NO POWER). EUC17 re-read the lens rows with a mask-aperture stellar mass and is NOT EVALUABLE as frozen, through a design error in one check.

9.5 Sealed predictions

PRJ1 sealed four predictions of the candidate on 2026-11-13, in a commit that adds `SEALED_PREDICTIONS.md` and `SEAL.txt` alone (99948d2e1e; on origin 9f009f33a0), with sha256 5e15f8240a4ef39ce87f3d3838163c400ccdc656dc98c8d6f5c9db3fd19ee51e for the sealed file; every number in it is printed by a script committed before the seal, and the lead recomputed both hashes and re-ran that script. Sealing is not adoption, and the sealed files are not edited here. Rival R1 is the earlier split at FRZ1's constant $v_{\rm stiff}$ (265.8/288.1 km s^{-1}), rival R2 the unsplit settled family.

P1. **Cell A: massive discs at z 1.8 to 2.6 above the switch carry the record's reference halo, not the law.** It separates the candidate from R2 only. KILLED if the law beats the reference at $A = 1$ ($T \leq -9s^2$) or if the fitted multiple's 95 % interval excludes 1 on either side.

P2. **Cell B: massive discs at z 0.5 to 1.7 below the switch follow the law, not the reference.** It separates the candidate from R1 and from a standard halo at the record's amplitude. KILLED if $T \geq +9s^2$ or if a free halo beats the law by $9s^2$.

P3. **Cell C: moderate discs at z 3.5 to 6 above the switch carry the reference halo, not the law.** This is the one cell where the candidate differs from both rivals. KILLED as in P1.

P4. **Where the switch sits at z 3.5 to 6: on $\log M_b^*(z)$, not at $v_{\rm stiff}$ and not absent.** A fitted offset of the switch from $\log M_b^*(z)$ is zero; KILLED if its 95 % interval excludes 0. It depends on f on its law side and is run at $f = 1.25$ and 2.

A SURVIVES is a hit and a KILLED reading counts only after an independent blind re-grade; the two directions are treated alike. The bar on the data is strict, because the law and the reference differ by only 0.015 to 0.08 dex in circular speed at 3 to 5 $R_{1/2}$: a qualifying disc needs a rotation curve to at least 4 $R_{1/2}$ with at least two independent resolution elements beyond 3 $R_{1/2}$, beam-forward modelling with its own point-spread function and the points' covariance, a gas mass from a measured tracer with its conversion fixed before the fit, independent inclination, and the pressure correction as published; the seven curves DSC1 and DSC1R graded as digitised are excluded. A cell has power only with at least 5 qualifying discs and a separation of at least $18s^2$ on noise-free mocks; otherwise it reads NO POWER. No lane has graded any disc in cells C or P4. Existing data lean against the second prediction: on the resolved discs below the switch both builds prefer a free reference halo at A about 1.25 to 1.57 over the law, and that on-disk reading, mostly below R1's boundary and so outside cell B, leans toward a halo, against P2, and decides nothing. Not sealed, and recorded as such: the forest (a kill-only condition, not distinctive), the strong lenses (already registered as EUC1's gate, where the candidate's prediction is the law's), groups at $z \approx 0$, wide binaries and the Solar System (equal to the law), and massive local spirals (the outcome may already be public).

9.6 Every lane folded into version 1.3

Labels canonical/alt as each file states them; "both" means the same label on both footings. "awaits" means awaits an independent re-grade.

Lane	Question	Label as frozen	Post hoc / re-grade
CJ1	the completed form and its scorecard	an assembly; grades nothing; NOT ADOPTED	audit 116 of 116
JNT2	the crossing-speed joint	JOINT FAILS ON G2 (coverage) (both)	JNT2R, blind
JNT2R	JNT2, blind	JOINT FAILS ON G2 (coverage), FRAGILE (both)	
JNT1, JNT1R	the earlier split on massive SPARC spirals	JNT1 NO VERDICT / JOINT COMPLETE ON POWERED ROWS [1.25, 2]; JNT1R REFERENCE FAILS AT THE RECORD SHARE, FRAGILE (canonical)	JNT1R not fully blind (its own file)
SRJ1–SRJ4	settled joint on clusters; the regime split; the reference layout on RC100 discs	SRJ1 JOINT FAILS ON CLUSTERS, FRAGILE; SRJ2 JOINT FAILS ON G2, G3, G4 / JOINT FAILS ON G2, G4, FRAGILE; SRJ3, SRJ4 NO VERDICT (no power)	
SRF1, SRF1R	the settled family at flux level	NO VERDICT / NO VERDICT (both builds)	SRF1R blind
FRZ4	the forest graded at flux level	full layout FOREST ROW FAILS AT FLUX LEVEL (both); settled NO VERDICT	no amendment
FRZ5	the forest's gas parameters refit	NO VERDICT (a twin does not bite) (both)	A1: FOREST ROW FAILS AFTER REFIT (both)
FRZ6	the split at flux level	HOLDS at $f = 2$ (canonical), HOLDS, FRAGILE ($f = 2$ alt; $f = 3$ both); control NO VERDICT	A1: control FAILS; a hit; FRZ6R does not confirm the label and does not contradict it
FRZ6R	FRZ6, blind	NO VERDICT for every split case (both); $f = 3$ FRAGILE (lambda_P x 0.5); control FAILS	A1: a reported bite rule, same words; TW1 does not bite
DSC1	the joint on resolved discs	above HOLDS, FRAGILE (canonical) / NO POWER; below LAW FAILS BELOW, FRAGILE	DSC1R, blind, confirms neither
DSC1R	DSC1, blind	above NO POWER, FRAGILE; below LAW HOLDS BELOW, FRAGILE (both)	RECONCILE: points counted as independent; the decision line

Lane	Question	Label as frozen	Post hoc / re-grade
ALP1, ALP1R	ALPAKA ring curves	NO POWER / NO POWER; ALP1R NO VERDICT, reading NO POWER	ALP1R blind
LNJ1	the switch on Euclid lenses	NO POWER, FRAGILE / NO POWER	
EUC11, EUC14, EUC15	source redshifts for EUC1's gate	EUC11, EUC14 NOT EVALUABLE; EUC15 EARLY READING, NOT A GATE	
EUC16	better arc finder	NO VERDICT / NO VERDICT; the gate graded for the first time	
EUC17	mask-aperture stellar mass	NOT EVALUABLE (a check's design error)	A2, post hoc
EXR1R, EXR2	the wide-lens cusp in X-rays	NOT EVALUABLE / NOT EVALUABLE as frozen, as EXR1	EXR1R blind
MWT1	what the hot bin rests on	FAIL DEPENDS ON AN ASSUMPTION THE DATA CANNOT TEST (census, both); rc alt FAIL REMOVED, FRAGILE	no amendment; awaits
MWT2	is the hot bin a structure	NO POWER (every case)	A1 post hoc, same word
RRL1	the slope from RR Lyrae	MEASURED SLOPE MAKES A PASSING BIN FAIL [40-55 COLD] (every case; FRAGILE in three)	no amendment
RFX1	the LMC reflex	REFLEX WITHIN THE BOUND, FRAGILE	
DPR1, KAP1	distinctive predictions; κ	DPR1 none dated for form B as adopted; KAPPA NOT DERIVED	
PRJ1	sealed predictions	four statements sealed; NOT ADOPTED	sealed files untouched
NUF1	the neutrino floor, flat ΛCDM's geometry	FLOOR DISFAVOURED / FRAGILE	grades nothing of this framework

9.7 Where this leaves the form

Version 1.3 changes no label of an earlier lane. The record now has a candidate for its sixth rule that adds no constant, places every galaxy at $z = 0$, every KiDS lens and the groups on the law and every X-COP cluster on the reference, and removes the earlier split's problem with massive SPARC spirals. As frozen it fails on one line, the coverage of massive discs at $z \approx 2$, where the data have no power; its passes are trivial or the law's; its one non-trivial data hit on resolved discs was not confirmed by a blind build; and its forest hit at flux level was neither confirmed as a label nor contradicted by its blind re-grade. Form B's failing rows stay, the joint is a rule and not an equation, and κ and 5.41 are fitted. The candidate is NOT ADOPTED. The theory is not closed. Two lanes were still running when version 1.3 was written and are not in it: FWN1, the split at $f = 5$, 10 and infinity at flux level, and WLJ1, the matter-power and weak-lensing rows with the split.

10 Our own estimates

Recorded before the numbers that tested them. (i) The KO2 brief put the error on the disc ratio near 0.2 and expected discs low and early types at 0.7 to 1.0. A model-only toy at the freeze gave the brief's ratio about twice the error of its replacement (0.37 against 0.19), so the lane replaced the primary statistic before computing; the brief's ratio then measured an error of 0.278 on discs, and discs came out undecided. (ii) The expectation that discs carry no outer field rested on the host-free lead, which does not generalise to the full sample. (iii) The RB1 brief expected X-COP near 0.96; it reads 1.095 on the forward-fitted mass. (iv) RB2 expected M31's bright dwarfs to be borderline; they fail. (v) The Bullet estimate, closer than 2.6σ once the stripped gas is counted, held; so did the group range and the Coma shift. (vi) CO2's hand estimate, written into its criteria before any code, put the optical depth at 0.055 to 0.060, σ_8 near 0.828, PR4 lensing within 1σ and raw KiDS near the line between tension and exclusion. The optical depth came out just below that range (0.0532); the rest held. (vii) The lead session and TPA1 expected the instability to survive the coupling to gravity, and the draft of this note carried it as the static state's own. It did not survive: SCS1's expectation before its freeze, stable with the earlier result planar, is the outcome. Version 1.1: (viii) The closure session expected the survey's newer release to cut the weak-lensing gap from about five widths to one or two, and the neutrino floor, the covariance and the gas together to be worth about one more. The release cut the σ_8 gap by about 45 % and left 4.4/4.6 of Legacy's narrower widths; the three inputs were worth 2.3 more widths on Legacy and 1.7 on KiDS-1000. (ix) It expected gas from the stellar-mass-selected stack to be worth 0.5 to 1 widths; it was worth 0.05 to 0.09 and only a bracket. (x) It proposed that the dwarfs that sit high are the ones that lost their gas. DG1 found the class medians

matched and the per-galaxy gates failed. (xi) It proposed the acceleration key and wrote that its gradient term had a healthy sign. AKD1 found the sign the other way: the term is kinetic, with a negative coefficient. Version 1.2: (xii) FRZ1's hand estimate expected the forest row to fail, possibly FRAGILE; the redistribution came out with the opposite sign and ten times the estimated size at k 3 to $5\,h\,\mathrm{Mpc}^{-1}$, and the fail is not FRAGILE. (xiii) FRZ1R's hand estimate, written blind, also had the sign wrong (+0.02 to +0.10 expected). (xiv) BX1's estimate before its second pair called GAP STAYS on DES Y3 most likely; it was not reached, and the measured noise came out below the estimated range. (xv) ELA1's hand estimate had a sign wrong that changes the outcome of the gate expected most likely to kill; its amendment, post hoc, corrects it. (xvi) PLT1 expected the halo stars and the satellites to split; they did not. (xvii) FALSIFIER_MATRIX row 32 registered, with DW1, a plateau prediction near 100 kpc and a kill at a measured $M(< 200\,\mathrm{kpc}) \geq 1.0 \times 10^{12}\,M_{\odot}$. The plateau was a product of the earlier reading of the external field; under the reading the owner adopted on 2026-11-12 the framework's own line would fire that kill, and the owner retired it the same day, after the data were seen. The row keeps only its re-scoped slope kill. (xviii) FRZ3 took its pressure-smoothing scale from memory as 80 proper kpc; the published value, as FRZ3R and the lead read the source (PROVISIONAL), is 79 comoving kpc, about four times smaller. FRZ3's softening factor of 0.0010/0.0006 rested on it; the blind build at the published scale finds 0.072/0.097. (xix) FRZ3R's hand estimate put the flux $\Delta\chi^2$ at 20 to 300 and the softening at 0.003 to 0.03; the flux kept about three times more of the matter-level signal than estimated. Version 1.3: (xx) FRZ6R's hand estimate, written blind, expected the split to fail the forest at $f = 3$ and $f = 2$; it found values 50 to 10,000 times below the estimate. It had expected haloes just under the crossing speed at z 2.2 to 2.6 to keep much of the outskirt change; they keep a little, and that is the whole of D.

Method

Lanes KO2, RB1, RB2, CO1 and CO2 are under `closure_2026_11_09/`; HE1, ONE1, TPA1, SCS1 and the notes FB1 and KO3F are under `fable52_push/`. None is registered in the closure record and none relabels an earlier lane. Post hoc items are marked in their lanes: RB1's amendment on the cluster shape gate, KO2's four amendments (the third, form B's prediction, was committed before any fitted amplitude was opened), KO2's mutation twin that did not bite, CO1's restated mutation arm, CO2's tolerance amendment, CO2's realisation check, which was made after grading, and SCS1's three amendments. RB2 has no amendment. This note claims no Lean theorem of its own; the Lean build that earlier lanes compiled with was removed from the machine on the evening of 2026-11-09 and a build inside the repository has been used since 22:17 that evening, so a Lean file's status is what its lane's compile output says. Distances for the KiDS lenses use the record's fiducial flat background (Ω_m 0.3, H_0 70), as in the staged sums. No halo model, no concentration relation and no simulation fit enters a graded row of KO2, RB1 or RB2; one externally calibrated bracket in RB1 is labelled as such. CO1's redistribution rows, which CO2's weak-lensing row reuses, inherit a mass function, a bias and a concentration relation calibrated on simulations, and say so. CO2's low-multipole table is Planck 2018 and its high-multipole spectra are PR4. The owner's decision is recorded in `closure_2026_11_09/OWNER_DECISION_2026-11-09_FINAL_FORM_B.md`. The supplement holds every lane file.

Version 1.1's lanes: SCS2, SCS3, DCN1, MK1, MK2, COL1, FLD1, REL1 and SHP1 are under `fable52_push/`; TK1, AKD1, LFS1, CLS1, KL1, KL1R, DG1, NH2, XCV1, GO1, BX1, DSB1, PAR1, WV1, SMF1, the note R541 and its lanes R541A to R541E are under `closure_2026_11_10/`, beside the notes `THEORY_ASSEMBLED_2026-11-10.md` and `MEDIUM_DESIGN_2026-11-10.md`, which are assemblies and grade nothing. None is registered and none relabels an earlier lane. Post hoc items are marked in their lanes: KL1's amendments A1, C1, C2 and C3, on which its readings other than NO VERDICT rest; DG1's amendment; TK1's, FLD1's and REL1's amendments; COL1's four amendments; MK2's second, third and fourth amendments; SCS2's second to fourth, SCS3's four and DCN1's six; GO1's amendment; PAR1's two amendments, WV1's two and SMF1's one; DSB1's amendment, made after its freeze and before any graded number; BX1's amendment; SHP1's check made after grading; and the readings of NH2 and XCV1 other than NO VERDICT. WM1's seeding gate was added after its first result. AKD1 has no amendment; its extra readings were added after the freeze and set no label. The earlier lanes named in Section 7's survey of media (M5, PH1, IJ1, HF1 and TM1 under `rebuild_2026_10_22/`, QX28, FA3 and WM1 under the closure folders of 2026-11-06, 11-07 and 11-09) are quoted by the first lines of their result files only; those files are added to the supplement. The KiDS-Legacy release was fetched on the owner's go and sits beside the repository, not in it.

Version 1.2's lanes: CLS2, EDG1, EFN1, FRZ1, FRZ1R, FRZ2, FZS1, HLF1, SET1, SRH1, SRH1R and VSH1

are under `closure_2026_11_10/`; DW1, FINAL-12, FINAL15, FINAL16, FINAL18, FINAL2, FINAL4, FINAL6 and VCC1 under `closure_2026_11_11/`; AZ2, AZ2R, CLOSE2, EUC1, EUC2, EUC4, EUC4R, EUC5, FINAL7F, GO2, LWB1, LWB1R, MWG1, MWJ1, NMB1, OUT4, PLT1, U4L1, EFC1, EUC3, EUC6, EUC7, HG2, LIT1, MWI1, NIM2, RCF1, RG1, SAT1, SLK1, STM1, PIK2, EUC5R, EUC6R, EUC8, RAW1, FRZ3, FRZ3R, EUC9, EUC10, EUC12, EUC13, EUC13R, ESZ1, EXR1, HOT1 and LMC1 under `closure_2026_11_12/`; A0Z1, FRM1, NBC1 and PIK1 under `fable52_push/`; CLOSE-1, CRACK-1, ELA1, FINAL-1, FINAL-7, FINAL-10, FINAL-11, FINAL-14, FINAL-17, FINAL-19, FINAL5, GAS1, OUT-2, both OUT3 lanes and OUT5 under `hy4_preview_push/`; STR1 under `hy4_push/`; AKS1, VSJ1, TOE0, QG1, HZ1, NPJ1 and LSF1 under `toe_2026_11_10/`. The audit quoted beside them is `reviews/hy4_codex_audit_2026_11_12/` (A1 to A4). None is registered and none relabels an earlier lane. Post hoc items are marked in their lanes: HLF1's A1; FZS1's rule and SRH1's family, both chosen after FRZ1 was seen, and SRH1's A1; SRH1R's A1; VSH1's and VSJ1's A1; AKS1's Part T and its A1; EFN1's A1 to A4; CLOSE2's amendments; FINAL7F's A1; FINAL15's correction of defect D1; FINAL-17's A1; OUT3's A1; ELA1's A1; CRACK-1's A1; BX1's A1 and its addendum; LWB1's A1; AZ2's A1 and A2; EUC2's and EUC5's A1 and EUC5's calibration; U4L1's A1; PLT1's post hoc reading; EFC1, a consolidation prompted by LIT1; EUC3's diagnostics; RCF1's, SAT1's and STM1's A1; EUC5R's F1S; RAW1's A1 and A2, each written after a failed run and before the run it governs; FRZ3R's reading T, added before any flux number was opened; EUC9's A1 and its addendum; EUC13's A1; EXR1's A1; LMC1's A1 to A3, A3 written after its values were seen; MWJ1, a consolidation made with every input seen; TOE0's and QG1's amendments; and HZ1's added twin. The forest lanes rest on the record's inherited mass-function and concentration ingredients, and FRZ1's lead did not check their scalings against their sources.

Version 1.3's lanes: DPR1, EUC11, EUC14, EUC15, EXR1R, EXR2, KAP1, RFX1, SRF1, SRF1R and SRJ1 to SRJ4 are under `closure_2026_11_12/`, with the map DR4MAP that row 38 of CJ1's scorecard quotes; CJ1, JNT1, JNT1R, JNT2, JNT2R, FRZ4, FRZ5, FRZ6, DSC1, DSC1R, ALP1, ALP1R, LNJ1, EUC16, EUC17, MWT1, MWT2, RRL1, NUF1 and PRJ1 are under `closure_2026_11_13/`, with FRZ6's Zenodo receipts in `FRZ6Z_zenodo_record/`; route D's check C5, the source of the lemma, is `fable52_push/SHAPE_search/route_D_dark_energy_column.md`. None is registered and none relabels an earlier lane; CJ1 and PRJ1 grade nothing. Post hoc items are marked in their lanes: the settled-radius family, chosen after FRZ1; the crossing-speed boundary, post hoc to SRJ2 and JNT1; FRZ5's A1; FRZ6's A1, on which its control's FAILS and its unsplit holds rest; DSC1R's A1 and its reconciliation, written after its blind work was committed; MWT2's A1; EUC17's A2; and the lead's placements of the words for the completed form in CJ1's scorecard. FRZ6's 44 part files (49 MB) are not copied into the supplement; they are in the Zenodo record DOI 10.5281/zenodo.24210556 (`FRZ6_parts.zip`). FRZ6R, the blind re-grade of FRZ6, is under `closure_2026_11_13/`; its amendment A1, committed alone before any graded number, adds a relative bite rule for one twin as a reported reading. Two further lanes were running when version 1.3 was written and are not in it: FWN1 (the split at $f = 5$, 10 and infinity at flux level) and WLJ1 (the matter-power and weak-lensing rows with the split). The paragraph on prior work in Section 9.1 rests on a provisional web search, not on a committed file.